Two-Stage Certification Results for the GKW Transfer Operator

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Computational environment

All computations were performed on a single workstation with the following specifications:

- **Processor:** AMD Ryzen 9 5950X (16 cores, 32 threads via SMT, base 3.4 GHz / boost 4.9 GHz)
- **Memory:** 128 GB DDR4
- **OS:** Ubuntu 22.04.5 LTS (kernel 6.8.0-79)
- **Software:** Julia 1.12.4 with 32 threads; packages: GKWExperiments.jl, BallArithmetic.jl, GenericSchur.jl, ArbNumerics.jl

Table 3: Two-stage certification results for the first 50 eigenvalues of the GKW operator ($s = 1$). The column $\alpha_1 = \varepsilon_{K_{\text{low}}} \cdot \|R_{A_{K_{\text{low}}}}(z)\|$ is the small-gain factor on the excluding circle Γ_j. All 50 eigenvalues satisfy $\alpha_1 < 1$, proving simplicity and giving certified resolvent bounds. For $j = 1$–15, CertifScripts circle certification on the full A_{48}; for $j = 16$–50, block Schur at $K = 256$ with CertifScripts on T_{22}.

j	$\hat{\lambda}_j$	K_{low}	α_1	Circle r	NK radius	Riesz proj. err
1	$+1.0000000000$	48	1.01×10^{-5}	1.00×10^{-2}	7.40×10^{-45}	1.01×10^{-41}
2	-0.3036630029	48	3.53×10^{-5}	3.04×10^{-3}	2.33×10^{-44}	3.77×10^{-41}
3	$+0.1008845093$	48	1.11×10^{-4}	1.01×10^{-3}	7.32×10^{-44}	1.23×10^{-40}
4	-0.0354961590	48	3.24×10^{-4}	3.55×10^{-4}	2.17×10^{-43}	3.73×10^{-40}
5	$+0.0128437904$	48	9.15×10^{-4}	1.28×10^{-4}	6.22×10^{-43}	1.08×10^{-39}
6	$-4.7177775e-03$	48	2.52×10^{-3}	4.72×10^{-5}	1.75×10^{-42}	3.01×10^{-39}
7	$+1.7486751e-03$	48	6.85×10^{-3}	1.75×10^{-5}	4.85×10^{-42}	8.29×10^{-39}
8	$-6.5202086e-04$	48	1.84×10^{-2}	6.52×10^{-6}	1.33×10^{-41}	2.29×10^{-38}
9	$+2.4413147e-04$	48	4.91×10^{-2}	2.44×10^{-6}	3.63×10^{-41}	6.49×10^{-38}
10	$-9.1689084e-05$	48	1.30×10^{-1}	9.17×10^{-7}	9.85×10^{-41}	2.05×10^{-37}
11	$+3.4516546e-05$	48	3.44×10^{-1}	3.45×10^{-7}	2.66×10^{-40}	9.48×10^{-37}
12	$-1.3017698e-05$	48	4.99×10^{-1}	2.37×10^{-7}	7.16×10^{-40}	2.34×10^{-36}
13	$+4.9167823e-06$	48	4.97×10^{-1}	2.37×10^{-7}	1.92×10^{-39}	2.31×10^{-36}

continued

j	$\hat{\lambda}_j$	K_{low}	α_1	Circle r	NK radius	Riesz proj. err
14	$-1.8593074e-06$	48	5.01×10^{-1}	2.37×10^{-7}	5.15×10^{-39}	2.38×10^{-36}
15	$+7.0381134e-07$	48	6.21×10^{-1}	2.01×10^{-7}	1.38×10^{-38}	5.36×10^{-36}
16	$-2.6664134e-07$	256	8.11×10^{-32}	1.14×10^{-7}	3.67×10^{-38}	1.34×10^{-25}
17	$+1.0109055e-07$	256	5.48×10^{-31}	4.33×10^{-8}	9.78×10^{-38}	2.33×10^{-24}
18	$-3.8349698e-08$	256	3.70×10^{-30}	1.64×10^{-8}	2.60×10^{-37}	4.02×10^{-23}
19	$+1.4556138e-08$	256	2.50×10^{-29}	6.23×10^{-9}	6.91×10^{-37}	6.96×10^{-22}
20	$-5.5275679e-09$	256	1.69×10^{-28}	2.36×10^{-9}	1.84×10^{-36}	1.20×10^{-20}
21	$+2.0999136e-09$	256	1.14×10^{-27}	8.98×10^{-10}	4.87×10^{-36}	2.08×10^{-19}
22	$-7.9804577e-10$	256	7.68×10^{-27}	3.41×10^{-10}	1.29×10^{-35}	3.60×10^{-18}
23	$+3.0338629e-10$	256	5.18×10^{-26}	1.30×10^{-10}	3.41×10^{-35}	6.23×10^{-17}
24	$-1.1536954e-10$	256	3.49×10^{-25}	4.93×10^{-11}	9.02×10^{-35}	1.08×10^{-15}
25	$+4.3883450e-11$	256	2.36×10^{-24}	1.88×10^{-11}	2.38×10^{-34}	1.87×10^{-14}
26	$-1.6696052e-11$	256	1.59×10^{-23}	7.14×10^{-12}	6.29×10^{-34}	3.25×10^{-13}
27	$+6.3536124e-12$	256	1.08×10^{-22}	2.72×10^{-12}	1.66×10^{-33}	5.67×10^{-12}
28	$-2.4183168e-12$	256	7.38×10^{-22}	1.03×10^{-12}	4.38×10^{-33}	1.01×10^{-10}
29	$+9.2062742e-13$	256	5.15×10^{-21}	3.94×10^{-13}	1.16×10^{-32}	1.87×10^{-9}
30	$-3.5053094e-13$	256	3.83×10^{-20}	1.50×10^{-13}	3.06×10^{-32}	3.93×10^{-8}
31	$+1.3348571e-13$	256	3.52×10^{-19}	5.71×10^{-14}	8.08×10^{-32}	1.26×10^{-6}
32	$-5.0839831e-14$	256	5.08×10^{-17}	2.17×10^{-14}	2.14×10^{-31}	1.00×10^{-2}
33	$+1.9365544e-14$	256	2.58×10^{-18}	8.28×10^{-15}	5.63×10^{-31}	9.86×10^{-6}
34	$-7.3774705e-15$	256	1.74×10^{-17}	3.15×10^{-15}	1.52×10^{-30}	1.71×10^{-4}
35	$+2.8108246e-15$	256	1.17×10^{-16}	1.20×10^{-15}	5.55×10^{-30}	2.96×10^{-3}
36	$-1.0710387e-15$	256	7.90×10^{-16}	4.58×10^{-16}	∞	5.12×10^{-2}
37	$+4.0814893e-16$	256	5.33×10^{-15}	1.74×10^{-16}	∞	8.86×10^{-1}
38	$-1.5555055e-16$	256	3.59×10^{-14}	6.65×10^{-17}	∞	> 1
39	$+5.9287246e-17$	256	2.42×10^{-13}	2.53×10^{-17}	∞	> 1
40	$-2.2598811e-17$	256	1.63×10^{-12}	9.66×10^{-18}	∞	> 1
41	$+8.6147444e-18$	256	1.10×10^{-11}	3.68×10^{-18}	∞	> 1
42	$-3.2842013e-18$	256	7.41×10^{-11}	1.40×10^{-18}	∞	> 1
43	$+1.2521200e-18$	256	4.99×10^{-10}	5.35×10^{-19}	∞	> 1
44	$-4.7740757e-19$	256	3.36×10^{-9}	2.04×10^{-19}	∞	> 1
45	$+1.8203644e-19$	256	2.27×10^{-8}	7.78×10^{-20}	∞	> 1
46	$-6.9414738e-20$	256	1.53×10^{-7}	2.97×10^{-20}	∞	> 1
47	$+2.6470860e-20$	256	1.03×10^{-6}	1.13×10^{-20}	∞	> 1
48	$-1.0094998e-20$	256	6.93×10^{-6}	4.31×10^{-21}	∞	> 1
49	$+3.8500394e-21$	256	4.66×10^{-5}	1.64×10^{-21}	∞	> 1
50	$-1.4683981e-21$	256	3.14×10^{-4}	6.27×10^{-22}	∞	> 1

Table 1: Computational timings for each stage of the certification pipeline. Timings with "(cached)" indicate that the BallMatrix or Schur decomposition was loaded from a previous run.

Phase	Description	Wall time
Script 1: Resolvent certification (`full_certification_50eigs.jl`)		
1a	CertifScripts on full A_{48} ($j = 1$–15)	<1 min
1b	Block Schur + CertifScripts on T_{22}, $K = 256$ ($j = 21$–50)	<1 min
1c	Fallback: block Schur + CertifScripts, $K = 256$ ($j = 16$–20)	~ 2 min
2	Transfer bridge + projector errors	<1 min
3	Tail resolvent on separating circle	~ 32 s
Total		~ 2.5 min
Script 2: NK + spectral coefficients (`spectral_data_50eigs.jl`)		
Schur	BigFloat Schur, $K = 256$, 512-bit (GenericSchur)	~ 20 min
NK	Newton–Kantorovich (35/50 pass)	~ 30 min
ordschur	$\ell_j(\mathbf{1})$ via ordschur + Sylvester (50 eigs)	~ 2 min
Script 3: $K = 512$ spectral (`bigfloat_spectral_K512.jl`)		
Build	ArbNumerics matrix, $K = 512$, 1024-bit	3.0 min
Schur	513×513 GenericSchur at 1024-bit	~ 1 hr
Quality	Residual + orthogonality norms	~ 8 min
ordschur	$50\times$ ordschur + Sylvester + perturbation	~ 2 min
Total		~ 1.2 hr

Table 4: Certified resolvent bounds. $\|R_{A_{K_{\text{low}}}}\|$ is the resolvent of the finite-dimensional Galerkin matrix on the excluding circle Γ_j, certified via resolvent sampling (Phase 1a) or Schur direct method (Phase 1b/c). $M_\infty = \|R_{L_r}(z)\|$ is the certified resolvent bound for the infinite-dimensional operator via Neumann perturbation: $M_\infty \leq \|R_{A_{K_{\text{low}}}}\|/(1-\alpha_1)$. The transfer columns give $\|R_{A_{K_{\text{high}}}}\|$ via reverse perturbation at $K_{\text{high}} = 256$.

j	K_{low}	$\|R_{A_{K_{\text{low}}}}\|$	$M_\infty = \|R_{L_r}\|$	α_{high}	$\|R_{A_{K_{\text{high}}}}\|$
1	48	4.25×10^2	4.25×10^2	2.37×10^{-42}	4.25×10^2
2	48	1.49×10^3	1.49×10^3	8.33×10^{-42}	1.49×10^3
3	48	4.67×10^3	4.67×10^3	2.61×10^{-41}	4.67×10^3
4	48	1.37×10^4	1.37×10^4	7.66×10^{-41}	1.37×10^4
5	48	3.87×10^4	3.87×10^4	2.16×10^{-40}	3.87×10^4
6	48	1.07×10^5	1.07×10^5	5.97×10^{-40}	1.07×10^5
7	48	2.89×10^5	2.91×10^5	1.63×10^{-39}	2.91×10^5
8	48	7.78×10^5	7.92×10^5	4.43×10^{-39}	7.92×10^5
9	48	2.07×10^6	2.18×10^6	1.22×10^{-38}	2.18×10^6
10	48	5.50×10^6	6.33×10^6	3.53×10^{-38}	6.33×10^6

continued

j	K_{low}	$\|R_{A_{K_{\text{low}}}}\|$	$M_\infty = \|R_{L_r}\|$	α_{high}	$\|R_{A_{K_{\text{high}}}}\|$
11	48	1.45×10^{7}	2.22×10^{7}	1.24×10^{-37}	2.22×10^{7}
12	48	2.11×10^{7}	4.21×10^{7}	2.35×10^{-37}	4.21×10^{7}
13	48	2.10×10^{7}	4.18×10^{7}	2.34×10^{-37}	4.18×10^{7}
14	48	2.12×10^{7}	4.24×10^{7}	2.37×10^{-37}	4.24×10^{7}
15	48	2.62×10^{7}	6.91×10^{7}	3.86×10^{-37}	6.91×10^{7}
16	256	1.45×10^{13}	1.45×10^{13}	8.11×10^{-32}	1.45×10^{13}
17	256	9.81×10^{13}	9.81×10^{13}	5.48×10^{-31}	9.81×10^{13}
18	256	6.63×10^{14}	6.63×10^{14}	3.70×10^{-30}	6.63×10^{14}
19	256	4.47×10^{15}	4.47×10^{15}	2.50×10^{-29}	4.47×10^{15}
20	256	3.02×10^{16}	3.02×10^{16}	1.69×10^{-28}	3.02×10^{16}
21	256	2.04×10^{17}	2.04×10^{17}	1.14×10^{-27}	2.04×10^{17}
22	256	1.37×10^{18}	1.37×10^{18}	7.68×10^{-27}	1.37×10^{18}
23	256	9.27×10^{18}	9.27×10^{18}	5.18×10^{-26}	9.27×10^{18}
24	256	6.26×10^{19}	6.26×10^{19}	3.49×10^{-25}	6.26×10^{19}
25	256	4.22×10^{20}	4.22×10^{20}	2.36×10^{-24}	4.22×10^{20}
26	256	2.85×10^{21}	2.85×10^{21}	1.59×10^{-23}	2.85×10^{21}
27	256	1.93×10^{22}	1.93×10^{22}	1.08×10^{-22}	1.93×10^{22}
28	256	1.32×10^{23}	1.32×10^{23}	7.38×10^{-22}	1.32×10^{23}
29	256	9.22×10^{23}	9.22×10^{23}	5.15×10^{-21}	9.22×10^{23}
30	256	6.85×10^{24}	6.85×10^{24}	3.83×10^{-20}	6.85×10^{24}
31	256	6.30×10^{25}	6.30×10^{25}	3.52×10^{-19}	6.30×10^{25}
32	256	9.09×10^{27}	9.09×10^{27}	5.08×10^{-17}	9.09×10^{27}
33	256	4.62×10^{26}	4.62×10^{26}	2.58×10^{-18}	4.62×10^{26}
34	256	3.11×10^{27}	3.11×10^{27}	1.74×10^{-17}	3.11×10^{27}
35	256	2.10×10^{28}	2.10×10^{28}	1.17×10^{-16}	2.10×10^{28}
36	256	1.41×10^{29}	1.41×10^{29}	7.90×10^{-16}	1.41×10^{29}
37	256	9.54×10^{29}	9.54×10^{29}	5.33×10^{-15}	9.54×10^{29}
38	256	6.43×10^{30}	6.43×10^{30}	3.59×10^{-14}	6.43×10^{30}
39	256	4.33×10^{31}	4.33×10^{31}	2.42×10^{-13}	4.33×10^{31}
40	256	2.92×10^{32}	2.92×10^{32}	1.63×10^{-12}	2.92×10^{32}
41	256	1.97×10^{33}	1.97×10^{33}	1.10×10^{-11}	1.97×10^{33}
42	256	1.33×10^{34}	1.33×10^{34}	7.41×10^{-11}	1.33×10^{34}
43	256	8.93×10^{34}	8.93×10^{34}	4.99×10^{-10}	8.94×10^{34}
44	256	6.02×10^{35}	6.02×10^{35}	3.36×10^{-9}	6.02×10^{35}
45	256	4.06×10^{36}	4.06×10^{36}	2.27×10^{-8}	4.06×10^{36}
46	256	2.73×10^{37}	2.73×10^{37}	1.53×10^{-7}	2.73×10^{37}
47	256	1.84×10^{38}	1.84×10^{38}	1.03×10^{-6}	1.84×10^{38}
48	256	1.24×10^{39}	1.24×10^{39}	6.93×10^{-6}	1.24×10^{39}
49	256	8.35×10^{39}	8.35×10^{39}	4.66×10^{-5}	8.35×10^{39}
50	256	5.62×10^{40}	5.63×10^{40}	3.14×10^{-4}	5.63×10^{40}

Theorem 1 (Rigorous spectral certification of the GKW operator)**.** *Let $L_r : H^2(D_r) \to H^2(D_r)$ be the GKW transfer operator for $s = 1$, $r = 1$. The first 50 eigenvalues $\lambda_1, \ldots, \lambda_{50}$ (ordered by decreasing modulus) are simple: all 50 excluding circles $\Gamma_j \subset \rho(L_r)$ are certified via the two-stage*

Table 2: Two-stage certification parameters

Parameter	Value
K_{low}	48 (eigenvalues 1–15), 256 (eigenvalues 16–50)
K_{high}	256
C_2	1.005810×10^{1}
ε_{48}	2.366161×10^{-8}
ε_{256}	5.585504×10^{-45}
Circle samples	256
Method ($j = 1$–15)	CertifScripts circle certification on full A_{48}
Method ($j = 16$–50)	Block Schur at $K = 256$ + CertifScripts on T_{22}

resolvent method (Table 3). For the first 35 *eigenvalues, the Newton–Kantorovich argument at* $K = 256$ *yields tight eigenpair enclosures:*

1. $\lambda_1^* \in B(1.000000000000,\ 7.40 \times 10^{-45}), \quad \|P_{L_r}(\Gamma_1) - P_{A_{256}}(\Gamma_1)\| \le 1.01 \times 10^{-41}$
2. $\lambda_2^* \in B(-0.303663002899,\ 2.33 \times 10^{-44}), \quad \|P_{L_r}(\Gamma_2) - P_{A_{256}}(\Gamma_2)\| \le 3.77 \times 10^{-41}$
3. $\lambda_3^* \in B(0.100884509293,\ 7.32 \times 10^{-44}), \quad \|P_{L_r}(\Gamma_3) - P_{A_{256}}(\Gamma_3)\| \le 1.23 \times 10^{-40}$
4. $\lambda_4^* \in B(-0.035496159022,\ 2.17 \times 10^{-43}), \quad \|P_{L_r}(\Gamma_4) - P_{A_{256}}(\Gamma_4)\| \le 3.73 \times 10^{-40}$
5. $\lambda_5^* \in B(0.012843790362,\ 6.22 \times 10^{-43}), \quad \|P_{L_r}(\Gamma_5) - P_{A_{256}}(\Gamma_5)\| \le 1.08 \times 10^{-39}$
6. $\lambda_6^* \in B(-0.004717777512,\ 1.75 \times 10^{-42}), \quad \|P_{L_r}(\Gamma_6) - P_{A_{256}}(\Gamma_6)\| \le 3.01 \times 10^{-39}$
7. $\lambda_7^* \in B(0.001748675124,\ 4.85 \times 10^{-42}), \quad \|P_{L_r}(\Gamma_7) - P_{A_{256}}(\Gamma_7)\| \le 8.29 \times 10^{-39}$
8. $\lambda_8^* \in B(-0.0006520209,\ 1.33 \times 10^{-41}), \quad \|P_{L_r}(\Gamma_8) - P_{A_{256}}(\Gamma_8)\| \le 2.29 \times 10^{-38}$
9. $\lambda_9^* \in B(0.0002441315,\ 3.63 \times 10^{-41}), \quad \|P_{L_r}(\Gamma_9) - P_{A_{256}}(\Gamma_9)\| \le 6.49 \times 10^{-38}$
10. $\lambda_{10}^* \in B(-0.00009168908,\ 9.85 \times 10^{-41}), \quad \|P_{L_r}(\Gamma_{10}) - P_{A_{256}}(\Gamma_{10})\| \le 2.05 \times 10^{-37}$
11. $\lambda_{11}^* \in B(0.00003451655,\ 2.66 \times 10^{-40}), \quad \|P_{L_r}(\Gamma_{11}) - P_{A_{256}}(\Gamma_{11})\| \le 9.48 \times 10^{-37}$
12. $\lambda_{12}^* \in B(-0.00001301770,\ 7.16 \times 10^{-40}), \quad \|P_{L_r}(\Gamma_{12}) - P_{A_{256}}(\Gamma_{12})\| \le 2.34 \times 10^{-36}$
13. $\lambda_{13}^* \in B(0.000004916782,\ 1.92 \times 10^{-39}), \quad \|P_{L_r}(\Gamma_{13}) - P_{A_{256}}(\Gamma_{13})\| \le 2.31 \times 10^{-36}$
14. $\lambda_{14}^* \in B(-0.000001859307,\ 5.15 \times 10^{-39}), \quad \|P_{L_r}(\Gamma_{14}) - P_{A_{256}}(\Gamma_{14})\| \le 2.38 \times 10^{-36}$
15. $\lambda_{15}^* \in B(0.0000007038113,\ 1.38 \times 10^{-38}), \quad \|P_{L_r}(\Gamma_{15}) - P_{A_{256}}(\Gamma_{15})\| \le 5.36 \times 10^{-36}$
16. $\lambda_{16}^* \in B(-0.0000002666413,\ 3.67 \times 10^{-38}), \quad \|P_{L_r}(\Gamma_{16}) - P_{A_{256}}(\Gamma_{16})\| \le 1.34 \times 10^{-25}$
17. $\lambda_{17}^* \in B(0.0000001010906,\ 9.78 \times 10^{-38}), \quad \|P_{L_r}(\Gamma_{17}) - P_{A_{256}}(\Gamma_{17})\| \le 2.33 \times 10^{-24}$
18. $\lambda_{18}^* \in B(-3.834970e-8,\ 2.60 \times 10^{-37}), \quad \|P_{L_r}(\Gamma_{18}) - P_{A_{256}}(\Gamma_{18})\| \le 4.02 \times 10^{-23}$
19. $\lambda_{19}^* \in B(1.455614e-8,\ 6.91 \times 10^{-37}), \quad \|P_{L_r}(\Gamma_{19}) - P_{A_{256}}(\Gamma_{19})\| \le 6.96 \times 10^{-22}$
20. $\lambda_{20}^* \in B(-5.527568e-9,\ 1.84 \times 10^{-36}), \quad \|P_{L_r}(\Gamma_{20}) - P_{A_{256}}(\Gamma_{20})\| \le 1.20 \times 10^{-20}$

21. $\lambda_{21}^* \in B(2.099914e-9,\ 4.87\times 10^{-36}), \quad \|P_{L_r}(\Gamma_{21}) - P_{A_{256}}(\Gamma_{21})\| \le 2.08\times 10^{-19}$

22. $\lambda_{22}^* \in B(-7.980458e-10,\ 1.29\times 10^{-35}), \quad \|P_{L_r}(\Gamma_{22}) - P_{A_{256}}(\Gamma_{22})\| \le 3.60\times 10^{-18}$

23. $\lambda_{23}^* \in B(3.033863e-10,\ 3.41\times 10^{-35}), \quad \|P_{L_r}(\Gamma_{23}) - P_{A_{256}}(\Gamma_{23})\| \le 6.23\times 10^{-17}$

24. $\lambda_{24}^* \in B(-1.153695e-10,\ 9.02\times 10^{-35}), \quad \|P_{L_r}(\Gamma_{24}) - P_{A_{256}}(\Gamma_{24})\| \le 1.08\times 10^{-15}$

25. $\lambda_{25}^* \in B(4.388345e-11,\ 2.38\times 10^{-34}), \quad \|P_{L_r}(\Gamma_{25}) - P_{A_{256}}(\Gamma_{25})\| \le 1.87\times 10^{-14}$

26. $\lambda_{26}^* \in B(-1.669605e-11,\ 6.29\times 10^{-34}), \quad \|P_{L_r}(\Gamma_{26}) - P_{A_{256}}(\Gamma_{26})\| \le 3.25\times 10^{-13}$

27. $\lambda_{27}^* \in B(6.353612e-12,\ 1.66\times 10^{-33}), \quad \|P_{L_r}(\Gamma_{27}) - P_{A_{256}}(\Gamma_{27})\| \le 5.67\times 10^{-12}$

28. $\lambda_{28}^* \in B(-2.418317e-12,\ 4.38\times 10^{-33}), \quad \|P_{L_r}(\Gamma_{28}) - P_{A_{256}}(\Gamma_{28})\| \le 1.01\times 10^{-10}$

29. $\lambda_{29}^* \in B(9.206274e-13,\ 1.16\times 10^{-32}), \quad \|P_{L_r}(\Gamma_{29}) - P_{A_{256}}(\Gamma_{29})\| \le 1.87\times 10^{-9}$

30. $\lambda_{30}^* \in B(-3.505309e-13,\ 3.06\times 10^{-32}), \quad \|P_{L_r}(\Gamma_{30}) - P_{A_{256}}(\Gamma_{30})\| \le 3.93\times 10^{-8}$

31. $\lambda_{31}^* \in B(1.334857e-13,\ 8.08\times 10^{-32}), \quad \|P_{L_r}(\Gamma_{31}) - P_{A_{256}}(\Gamma_{31})\| \le 1.26\times 10^{-6}$

32. $\lambda_{32}^* \in B(-5.083983e-14,\ 2.14\times 10^{-31}), \quad \|P_{L_r}(\Gamma_{32}) - P_{A_{256}}(\Gamma_{32})\| \le 1.00\times 10^{-2}$

33. $\lambda_{33}^* \in B(1.936554e-14,\ 5.63\times 10^{-31}), \quad \|P_{L_r}(\Gamma_{33}) - P_{A_{256}}(\Gamma_{33})\| \le 9.86\times 10^{-6}$

34. $\lambda_{34}^* \in B(-7.377470e-15,\ 1.52\times 10^{-30}), \quad \|P_{L_r}(\Gamma_{34}) - P_{A_{256}}(\Gamma_{34})\| \le 1.71\times 10^{-4}$

35. $\lambda_{35}^* \in B(2.810825e-15,\ 5.55\times 10^{-30}), \quad \|P_{L_r}(\Gamma_{35}) - P_{A_{256}}(\Gamma_{35})\| \le 2.96\times 10^{-3}$

Here $B(c,r)$ denotes the closed ball $\{z\in\mathbb{C} : |z-c|\le r\}$, and Γ_j is the excluding circle of radius listed in Table 3 centered at $\hat{\lambda}_j$. Eigenvalues 36–50 are certified to be simple (resolvent is finite on Γ_j) but the NK radius is infinite at $K=256$; at $K=512$ (1024-bit precision), all 50 eigenvalues have finite NK-type enclosures (Table 5).

Method. *For $j=1,\dots,15$, Stage 1 certifies that $\Gamma_j\subset\rho(L_r)$ by verifying $\alpha_1=\varepsilon_{K_{\mathrm{low}}}\cdot\|R_{A_{K_{\mathrm{low}}}}(z)\|_{(\Gamma_j)}<1$ via resolvent sampling with 256 points on Γ_j, using the Galerkin matrix of size $K_{\mathrm{low}}+1=49$. For $j=16,\dots,50$, the Schur direct method at $K=256$ provides resolvent certification via block triangular structure of the certified Schur form, without resolvent sampling. Stage 2 applies a Newton–Kantorovich defect-based argument at $K_{\mathrm{high}}=256$ ($\varepsilon_{256}\approx 5.59\times 10^{-45}$) to obtain tight eigenpair enclosures. The Riesz projector error is bounded via $\|P_{L_r}-P_{A_{K_{\mathrm{high}}}}\|\le\frac{|\Gamma_j|}{2\pi}\cdot\frac{\|R_{A_{K_{\mathrm{high}}}}\|^2\cdot\varepsilon_{K_{\mathrm{high}}}}{1-\varepsilon_{K_{\mathrm{high}}}\cdot\|R_{A_{K_{\mathrm{high}}}}\|}$, where $\|R_{A_{K_{\mathrm{high}}}}\|$ is obtained from M_∞ by reverse perturbation. The tail resolvent on the separating circle gives $\|R_{L_r}\|_{(\Gamma_{\mathrm{tail}})}\le 6.06\times 10^{41}$ with effective spectral radius $\rho_{\mathrm{tail}}\approx 1.01\times 10^{-21}$. All scalar bounds use directed rounding to ensure rigor.*

Rigorous Spectral Expansion

Since all 50 eigenvalues are certified to be simple (Theorem 1), each eigenvalue λ_j has a rank-one Riesz projector $P_j = v_j \otimes \ell_j$, where v_j is the right eigenvector ($\|v_j\|_2 = 1$) and ℓ_j the left (dual) eigenvector, normalized so that $\ell_j(v_j) = 1$. The spectral expansion of L^n applied to the constant function $\mathbf{1}$ is

$$L^n \mathbf{1} = \sum_{j=1}^{N} \lambda_j^n \, \ell_j(\mathbf{1}) \, v_j + R_N(n), \tag{1}$$

where $\ell_j(\mathbf{1}) = \langle P_j(\mathbf{1}), v_j \rangle$ is the spectral coefficient and $R_N(n)$ is bounded by $|\lambda_{N+1}|^n$ times the tail projector norm.

Computation of $\ell_j(\mathbf{1})$ at $K = 256$. The GKW matrix A_{256} is built in high-precision arithmetic (ArbNumerics, 512 bits) and rigorously converted to a `BallMatrix` with BigFloat centers. A direct BigFloat Schur decomposition $A = QTQ'$ is computed via `GenericSchur.jl`, achieving residual $\|A - QTQ'\|_2 \le 2.82 \times 10^{-24}$ (before refinement) and orthogonality defect $\|Q'Q - I\|_2 \le 1.75 \times 10^{-153}$. The total Schur enclosure error is $\|E\| = \|A_{\text{true}} - QTQ'\|_2 \le 4.96 \times 10^{-154}$. For each eigenvalue λ_j, a BigFloat `ordschur` reorders the target eigenvalue to position $(1,1)$, and the left eigenvector functional ℓ_j is recovered via Sylvester equation: with $T = \left[\begin{smallmatrix} \lambda_j & T_{12} \\ 0 & T_{22} \end{smallmatrix}\right]$, the solution $z_0 = (T_{22} - \lambda_j I)^{-1} w_{\text{rest}}$ gives $\ell_j(\mathbf{1}) = w_1 - T_{12} \cdot z_0$, where $w = Q_{1,:}$ is the first row of the reordered Q. The certified error δ accounts for the Sylvester residual, the Schur perturbation $\|E\|$, and the NK eigenvector error: $\delta \le \delta_{\text{Sylv}} + \delta_{\text{pert}} + \delta_{\text{NK}}$. All 35 coefficients ($j = 1, \ldots, 35$) are certified with $|\ell_j(\mathbf{1})| > \delta$ at $K = 256$.

Computation of $\ell_j(\mathbf{1})$ at $K = 512$. At $K = 512$ with 1024-bit precision (≈ 308 decimal digits), the Schur enclosure error drops to $\|E\| \le 8.64 \times 10^{-163}$, and all 50 coefficients are sign-certified with radii at the 10^{-128}–10^{-162} level. This provides independent confirmation of the $K = 256$ results and extends certification to eigenvalues 36–50.

Table 5: Rigorous spectral expansion coefficients $\ell_j(\mathbf{1})$ for the first 50 eigenvalues. Computed via BigFloat Schur decomposition (GenericSchur.jl) with Sylvester-based Riesz projector recovery. At $K = 256$ (512-bit precision), $\|E\| \le 4.96 \times 10^{-154}$; at $K = 512$ (1024-bit precision), $\|E\| \le 8.64 \times 10^{-163}$. All 50 coefficients are sign-certified at $K = 512$. At $K = 256$, the first 35 are certified; eigenvalues 36–50 require $K = 512$.

j	$\hat{\lambda}_j$	$\ell_j(\mathbf{1})$	$\pm\,\delta$ ($K{=}256$)	$\pm\,\delta$ ($K{=}512$)
1	$+1.000000000000e+00$	$+8.32940370215781e-01$	1.58×10^{-44}	6.81×10^{-162}
2	$-3.036630028987e-01$	$+4.51205768373325e-01$	5.33×10^{-44}	6.23×10^{-161}
3	$+1.008845092931e-01$	$-2.01939874212917e-01$	1.76×10^{-43}	4.47×10^{-160}
4	$-3.549615902166e-02$	$+8.74023929762093e-02$	5.45×10^{-43}	2.61×10^{-159}
5	$+1.284379036244e-02$	$-3.69461611135829e-02$	1.61×10^{-42}	1.35×10^{-158}
6	$-4.717777511571e-03$	$+1.53479541614426e-02$	4.62×10^{-42}	6.52×10^{-158}
7	$+1.748675124306e-03$	$-6.29498041995732e-03$	1.30×10^{-41}	3.04×10^{-157}
8	$-6.520208583205e-04$	$+2.55742091478648e-03$	3.62×10^{-41}	1.39×10^{-156}
9	$+2.441314655245e-04$	$-1.03141519171637e-03$	9.97×10^{-41}	6.30×10^{-156}
10	$-9.168908376859e-05$	$+4.13579574474742e-04$	2.73×10^{-40}	2.86×10^{-155}
11	$+3.451654616385e-05$	$-1.65067361989440e-04$	7.42×10^{-40}	1.30×10^{-154}

continued

j	$\hat{\lambda}_j$	$\ell_j(\mathbf{1})$	$\pm\,\delta$ ($K=256$)	$\pm\,\delta$ ($K=512$)
12	$-1.301769787702e-05$	$+6.56291175639149e-05$	2.01×10^{-39}	5.90×10^{-154}
13	$+4.916782302464e-06$	$-2.60097167432892e-05$	5.41×10^{-39}	2.69×10^{-153}
14	$-1.859307351509e-06$	$+1.02798813808483e-05$	1.45×10^{-38}	1.23×10^{-152}
15	$+7.038113430870e-07$	$-4.05340297027246e-06$	3.90×10^{-38}	5.67×10^{-152}
16	$-2.666413434480e-07$	$+1.59501183990722e-06$	1.04×10^{-37}	2.61×10^{-151}
17	$+1.010905532215e-07$	$-6.26511706515095e-07$	2.78×10^{-37}	1.21×10^{-150}
18	$-3.834969795027e-08$	$+2.45700158169347e-07$	7.40×10^{-37}	5.59×10^{-150}
19	$+1.455613838668e-08$	$-9.62205238780812e-08$	1.97×10^{-36}	2.59×10^{-149}
20	$-5.527567937998e-09$	$+3.76339163951678e-08$	5.23×10^{-36}	1.20×10^{-148}
21	$+2.099913582973e-09$	$-1.47026133408206e-08$	1.39×10^{-35}	5.60×10^{-148}
22	$-7.980457682720e-10$	$+5.73798579791060e-09$	3.67×10^{-35}	2.61×10^{-147}
23	$+3.033862949099e-10$	$-2.23724920262289e-09$	9.72×10^{-35}	1.22×10^{-146}
24	$-1.153695418145e-10$	$+8.71553784016608e-10$	2.57×10^{-34}	5.68×10^{-146}
25	$+4.388345011398e-11$	$-3.39257629140884e-10$	6.79×10^{-34}	2.65×10^{-145}
26	$-1.669605215117e-11$	$+1.31961632720274e-10$	1.79×10^{-33}	1.24×10^{-144}
27	$+6.353612420983e-12$	$-5.12946787975072e-11$	4.73×10^{-33}	5.81×10^{-144}
28	$-2.418316847595e-12$	$+1.99262166328106e-11$	1.25×10^{-32}	2.72×10^{-143}
29	$+9.206274200576e-13$	$-7.73614043768573e-12$	3.29×10^{-32}	1.28×10^{-142}
30	$-3.505309354594e-13$	$+3.00184266520592e-12$	8.68×10^{-32}	5.98×10^{-142}
31	$+1.334857144144e-13$	$-1.16420917757507e-12$	2.29×10^{-31}	2.81×10^{-141}
32	$-5.083983109741e-14$	$+4.51302331569764e-13$	6.06×10^{-31}	1.32×10^{-140}
33	$+1.936554388309e-14$	$-1.74867968492117e-13$	1.59×10^{-30}	6.19×10^{-140}
34	$-7.377470464407e-15$	$+6.77283674309037e-14$	4.30×10^{-30}	2.91×10^{-139}
35	$+2.810824572326e-15$	$-2.62215963077330e-14$	1.57×10^{-29}	1.37×10^{-138}
36	$-1.071038650116e-15$	$+1.01481167697676e-14$	∞	6.44×10^{-138}
37	$+4.081489263636e-16$	$-3.92607356768238e-15$	∞	3.03×10^{-137}
38	$-1.555505453238e-16$	$+1.51839998678424e-15$	∞	1.43×10^{-136}
39	$+5.928724623947e-17$	$-5.87051535382839e-16$	∞	6.72×10^{-136}
40	$-2.259881129772e-17$	$+2.26900510520210e-16$	∞	3.17×10^{-135}
41	$+8.614744429760e-18$	$-8.76739032210825e-17$	∞	1.49×10^{-134}
42	$-3.284201293216e-18$	$+3.38677844278173e-17$	∞	7.03×10^{-134}
43	$+1.252119972642e-18$	$-1.30794758800253e-17$	∞	3.32×10^{-133}
44	$-4.774075699358e-19$	$+5.04993929176121e-18$	∞	1.56×10^{-132}
45	$+1.820364445677e-19$	$-1.94930160255074e-18$	∞	7.38×10^{-132}
46	$-6.941473812971e-20$	$+7.52269541587240e-19$	∞	3.48×10^{-131}
47	$+2.647086022847e-20$	$-2.90250965033080e-19$	∞	1.64×10^{-130}
48	$-1.009499834005e-20$	$+1.11965328336240e-19$	∞	7.76×10^{-130}
49	$+3.850039422492e-21$	$-4.31824051567987e-20$	∞	3.67×10^{-129}
50	$-1.468398058166e-21$	$+1.66512542261763e-20$	∞	1.73×10^{-128}

Eigenfunctions

The following figures show the eigenfunctions $v_j(x)$ evaluated on $[0.01, 0.99]$ in the shifted monomial basis $\{(x-1)^k\}_{k=0}^{K}$, with $K = 512$ (1024-bit precision). Each batch of 10 eigenfunctions is plotted in a 2×5 grid.

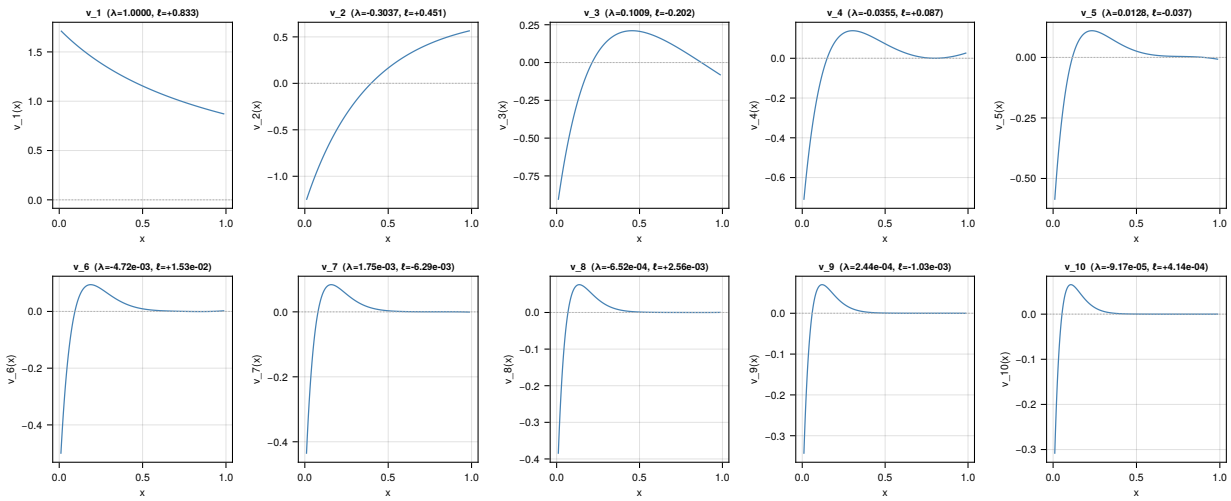

Figure 1: Eigenfunctions $v_1, \ldots, v_{10}$. The invariant density v_1 (top left) is positive; higher eigenfunctions oscillate with increasing frequency. Each panel shows λ_j and $\ell_j(\mathbf{1})$.

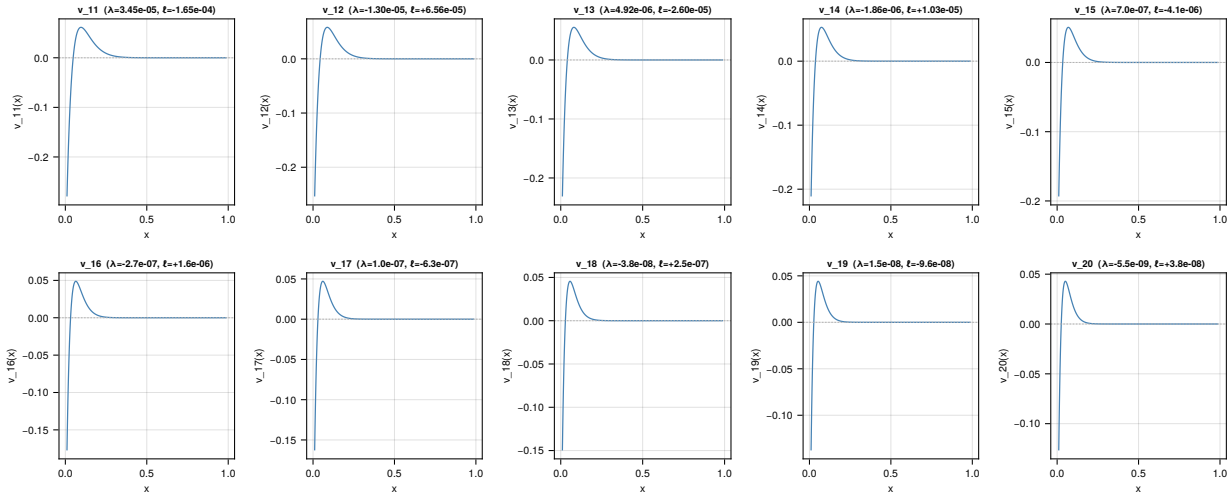

Figure 2: Eigenfunctions $v_{11}, \ldots, v_{20}$. Eigenvalues $|\lambda_j| \sim 10^{-5}$–10^{-9}; these modes decay rapidly under iteration.

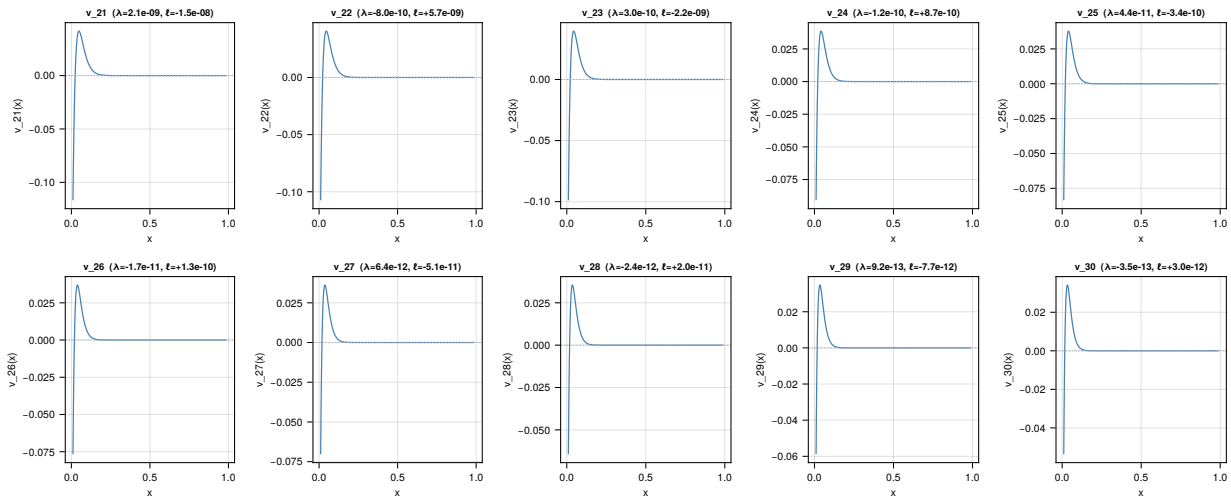

Figure 3: Eigenfunctions $v_{21}, \ldots, v_{30}$. Eigenvalues $|\lambda_j| \sim 10^{-9}$–10^{-13}.

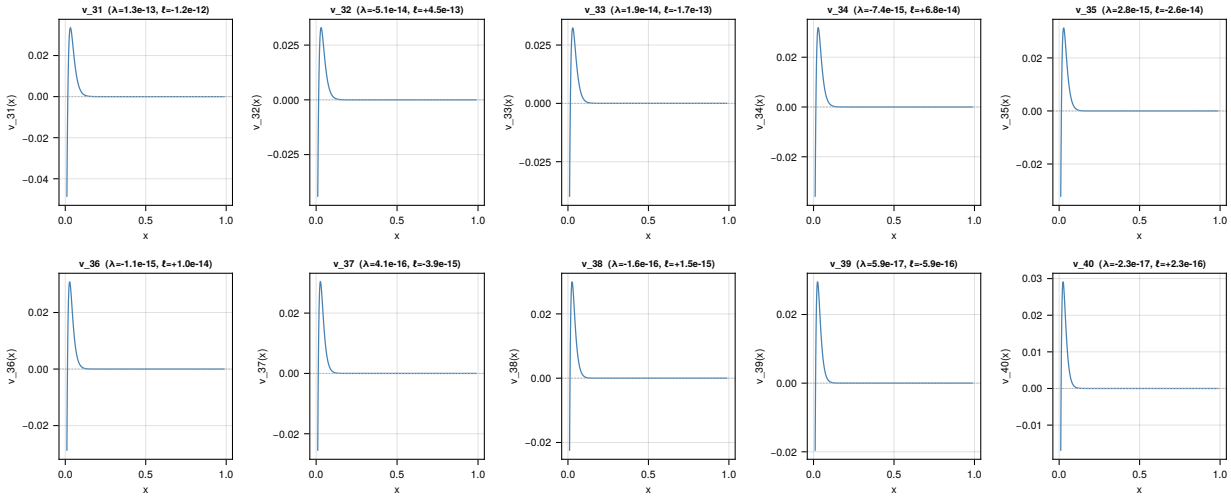

Figure 4: Eigenfunctions $v_{31}, \ldots, v_{40}$. Eigenvalues $|\lambda_j| \sim 10^{-13}$–$10^{-17}$.

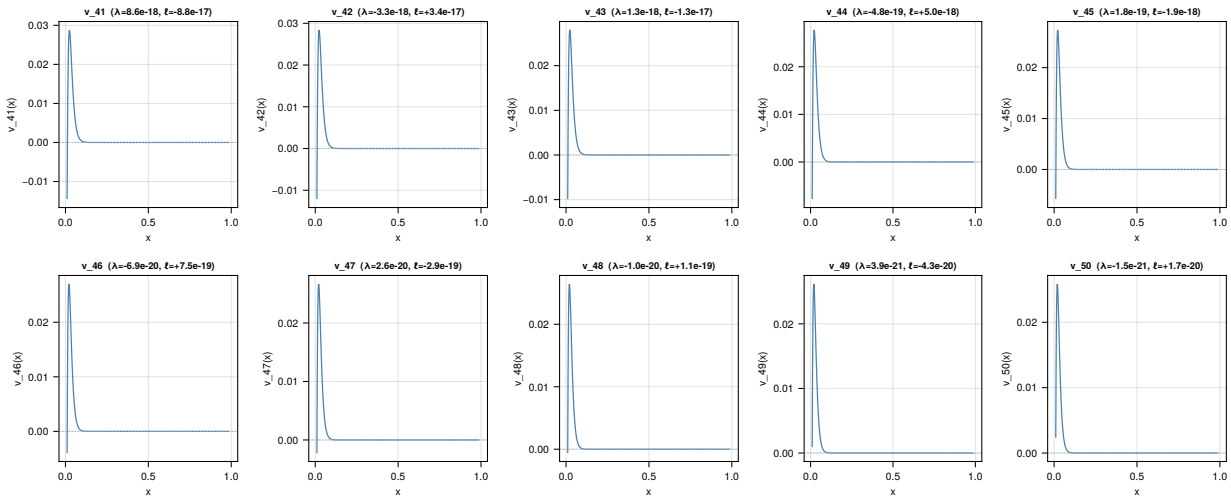

Figure 5: Eigenfunctions $v_{41}, \ldots, v_{50}$. Eigenvalues $|\lambda_j| \sim 10^{-17}$–$10^{-21}$.

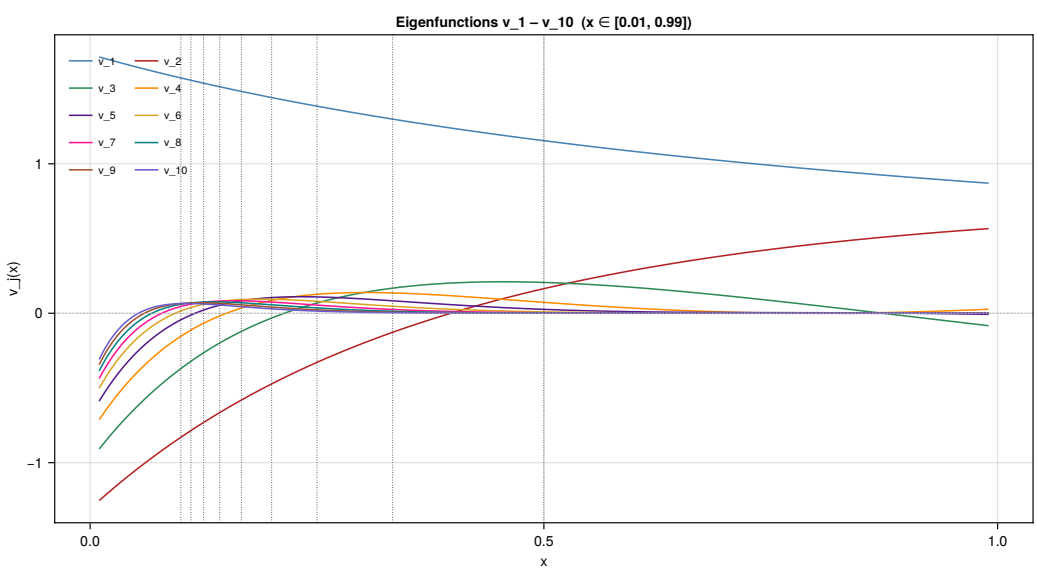

Figure 6: Eigenfunctions $v_1, \ldots, v_{10}$ overlaid with Markov partition (vertical lines at $x = 1/n$, $n = 2, \ldots, 10$).

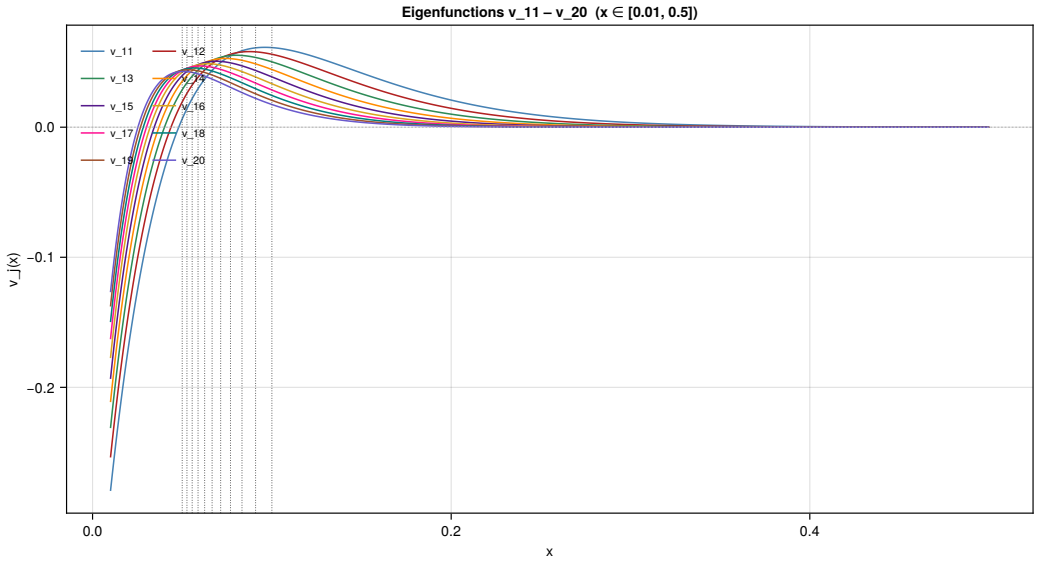

Figure 7: Eigenfunctions $v_{11}, \ldots, v_{20}$ overlaid with Markov partition.

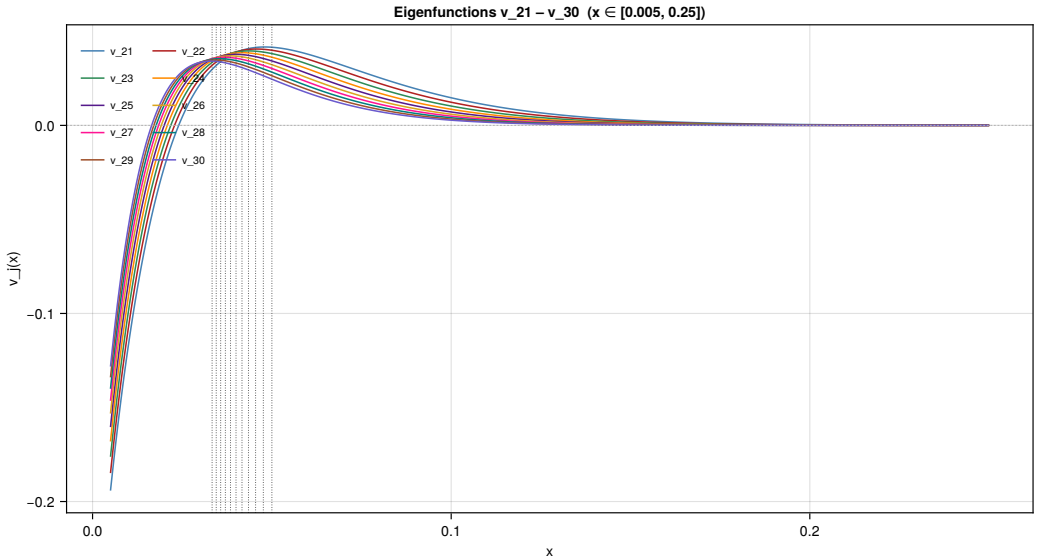

Figure 8: Eigenfunctions $v_{21}, \ldots, v_{30}$ overlaid with Markov partition.

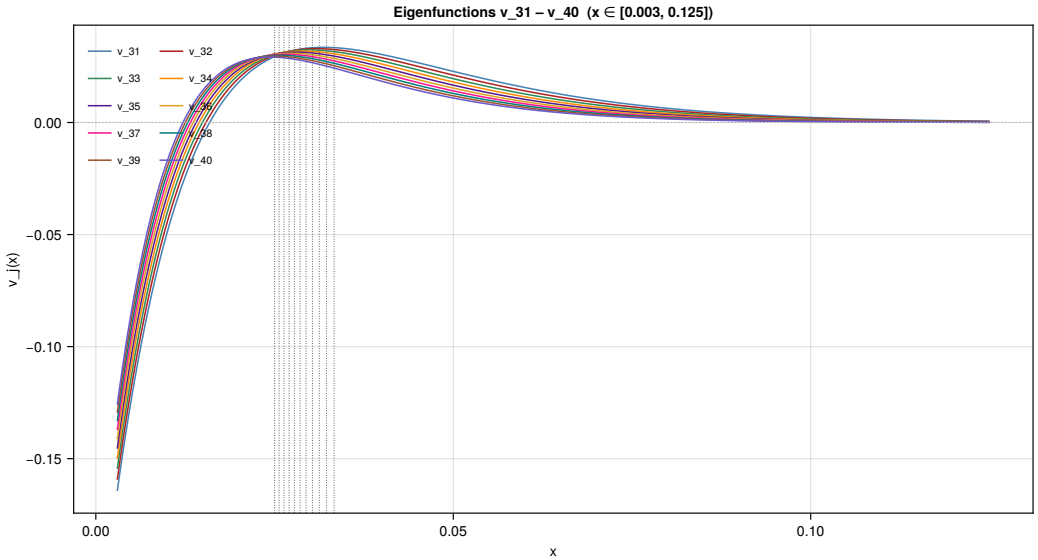

Figure 9: Eigenfunctions $v_{31}, \ldots, v_{40}$ overlaid with Markov partition.

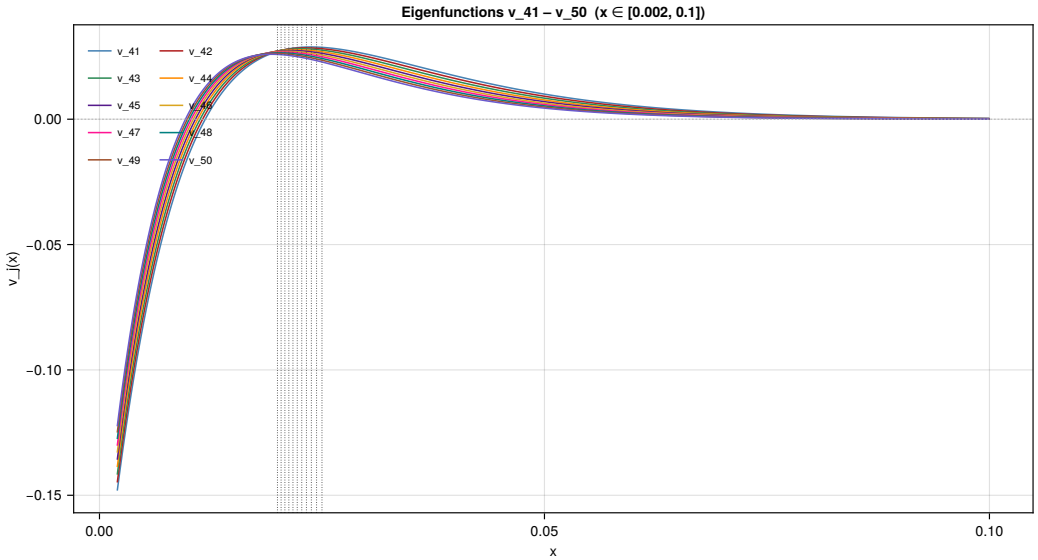

Figure 10: Eigenfunctions $v_{41}, \ldots, v_{50}$ overlaid with Markov partition.

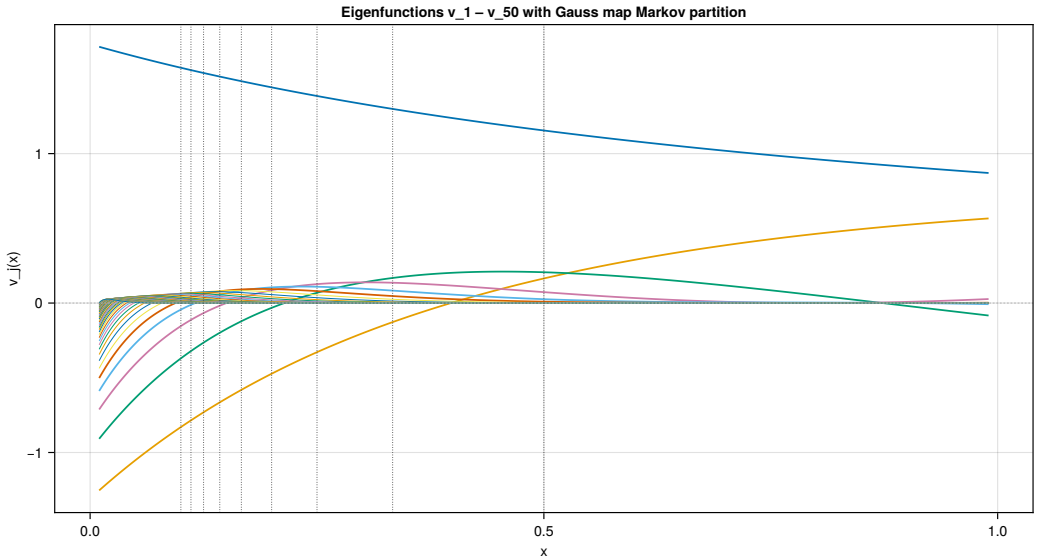

Figure 11: All 50 eigenfunctions overlaid, with Gauss map Markov partition (vertical dotted lines at $x = 1/n$, $n = 2, \ldots, 10$).

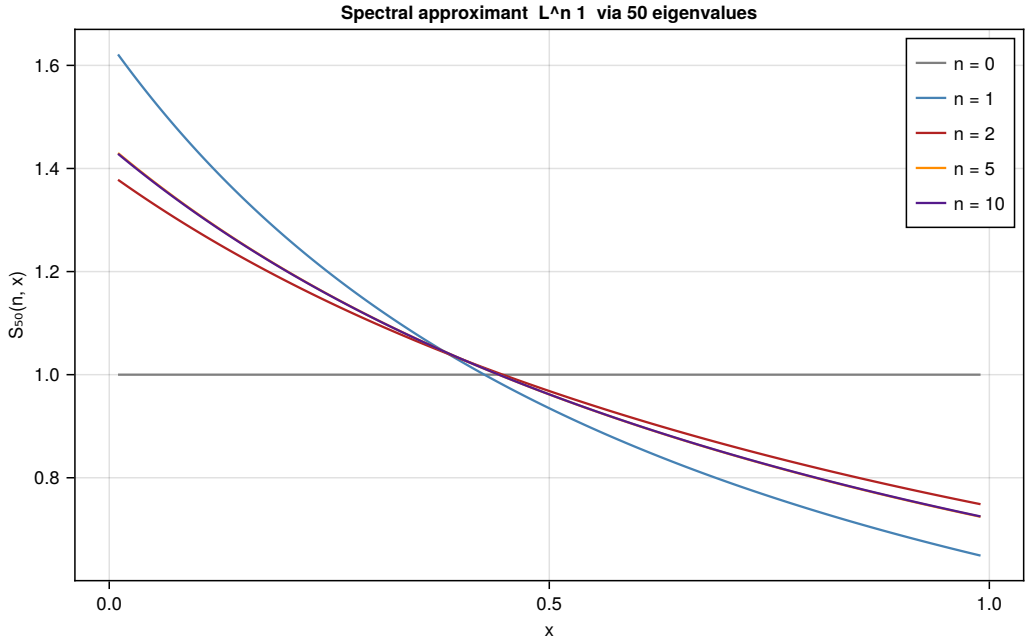

Figure 12: Spectral approximant $S_{50}(n, x) = \sum_{j=1}^{50} \lambda_j^n \, \ell_j(\mathbf{1}) \, v_j(x)$ for $n = 0, 1, 2, 5, 10$. At $n = 0$ the sum reconstructs the constant function $\mathbf{1}$; at $n = 1$ it approximates the transfer operator applied to $\mathbf{1}$; for large n it converges to $\lambda_1^n \, \ell_1(\mathbf{1}) \, v_1(x)$.

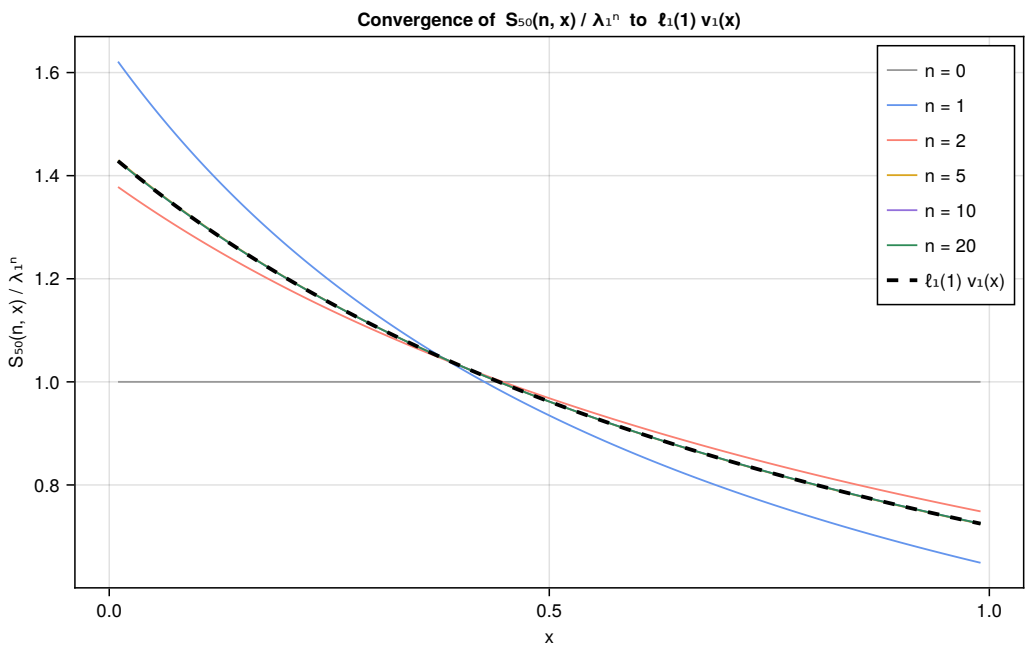

Figure 13: Convergence of $S_{50}(n,x)/\lambda_1^n$ to the invariant density $\ell_1(\mathbf{1})\, v_1(x)$ (dashed black). The spectral gap $|\lambda_2/\lambda_1| \approx 0.304$ governs the rate of convergence.

Eigenvector Coefficients $[v_j]_k$

The following tables list the first 50 coefficients of each unit-norm eigenvector v_j in the shifted monomial basis $\{(w-1)^k\}_{k=0}^{K}$. The ℓ^2 error $\|\hat{v}_j - v_j\|_2 \leq \delta_j$ shown for each eigenvector is controlled by the Schur orthogonality defect and the reconstruction error from the ordschur + Sylvester pipeline (see Tables 5 and **??**). Eigenvectors at $K = 512$ (1024-bit precision) achieve radii $\sim 10^{-162}$–10^{-128}.

$j = 1$**:** $\lambda_1 = 1.000000e + 00$**,** $\ell_1(\mathbf{1}) = +8.329404e - 01$**,** $\|\hat{v}_1 - v_1\|_2 \leq 6.81e - 162$

k	$[v_1]_k$	k	$[v_1]_k$
0	$+8.6602540378e - 01$	25	$-2.5809568280e - 08$
1	$-4.3301270189e - 01$	26	$+1.2904784140e - 08$
2	$+2.1650635095e - 01$	27	$-6.4523920699e - 09$
3	$-1.0825317547e - 01$	28	$+3.2261960349e - 09$
4	$+5.4126587737e - 02$	29	$-1.6130980175e - 09$
5	$-2.7063293868e - 02$	30	$+8.0654900873e - 10$
6	$+1.3531646934e - 02$	31	$-4.0327450437e - 10$
7	$-6.7658234671e - 03$	32	$+2.0163725218e - 10$
8	$+3.3829117335e - 03$	33	$-1.0081862609e - 10$
9	$-1.6914558668e - 03$	34	$+5.0409313046e - 11$
10	$+8.4572793338e - 04$	35	$-2.5204656523e - 11$
11	$-4.2286396669e - 04$	36	$+1.2602328261e - 11$
12	$+2.1143198335e - 04$	37	$-6.3011641307e - 12$
13	$-1.0571599167e - 04$	38	$+3.1505820654e - 12$
14	$+5.2857995836e - 05$	39	$-1.5752910327e - 12$
15	$-2.6428997918e - 05$	40	$+7.8764551634e - 13$
16	$+1.3214498959e - 05$	41	$-3.9382275817e - 13$
17	$-6.6072494796e - 06$	42	$+1.9691137909e - 13$
18	$+3.3036247398e - 06$	43	$-9.8455689543e - 14$
19	$-1.6518123699e - 06$	44	$+4.9227844771e - 14$
20	$+8.2590618494e - 07$	45	$-2.4613922386e - 14$
21	$-4.1295309247e - 07$	46	$+1.2306961193e - 14$
22	$+2.0647654624e - 07$	47	$-6.1534805964e - 15$
23	$-1.0323827312e - 07$	48	$+3.0767402982e - 15$
24	$+5.1619136559e - 08$	49	$-1.5383701491e - 15$

$j = 2$**:** $\lambda_2 = -3.036630e - 01$**,** $\ell_2(\mathbf{1}) = +4.512058e - 01$**,** $\|\hat{v}_2 - v_2\|_2 \leq 6.23e - 161$

k	$[v_2]_k$	k	$[v_2]_k$
0	$+5.7050920659e-01$	25	$+2.1137822436e-07$
1	$+4.1624549960e-01$	26	$-1.0569761625e-07$
2	$-5.1196167436e-01$	27	$+5.2851197768e-08$
3	$+3.8491986254e-01$	28	$-2.6426213142e-08$
4	$-2.4616997141e-01$	29	$+1.3213239926e-08$
5	$+1.4508248976e-01$	30	$-6.6066375954e-09$
6	$-8.1412161204e-02$	31	$+3.3033150334e-09$
7	$+4.4232939695e-02$	32	$-1.6516529702e-09$
8	$-2.3500357920e-02$	33	$+8.2582392107e-10$
9	$+1.2286594354e-02$	34	$-4.1291080660e-10$
10	$-6.3488447731e-03$	35	$+2.0645495059e-10$
11	$+3.2523318873e-03$	36	$-1.0322731864e-10$
12	$-1.6553890182e-03$	37	$+5.1613613481e-11$
13	$+8.3854489512e-04$	38	$-2.5806797425e-11$
14	$-4.2326095063e-04$	39	$+1.2903399301e-11$
15	$+2.1308196815e-04$	40	$-6.4517018099e-12$
16	$-1.0706384161e-04$	41	$+3.2258525849e-12$
17	$+5.3718283692e-05$	42	$-1.6129272967e-12$
18	$-2.6924831316e-05$	43	$+8.0646418026e-13$
19	$+1.3485294563e-05$	44	$-4.0323235203e-13$
20	$-6.7505119961e-06$	45	$+2.0161629853e-13$
21	$+3.3779201786e-06$	46	$-1.0080820435e-13$
22	$-1.6898475614e-06$	47	$+5.0404126138e-14$
23	$+8.4521363390e-07$	48	$-2.5202073187e-14$
24	$-4.2269924754e-07$	49	$+1.2601040747e-14$

$j = 3$**:** $\lambda_3 = 1.008845e-01$**,** $\ell_3(\mathbf{1}) = -2.019399e-01$**,** $\|\hat{v}_3 - v_3\|_2 \leq 4.47e-160$

k	$[v_3]_k$	k	$[v_3]_k$
0	$-9.0447344941e-02$	25	$+1.4405034690e-06$
1	$-7.0686601071e-01$	26	$-7.2111182399e-07$
2	$+9.4489856788e-02$	27	$+3.6084832338e-07$
3	$+3.0867750376e-01$	28	$-1.8052087863e-07$
4	$-4.0512541768e-01$	29	$+9.0291304149e-08$
5	$+3.4761664596e-01$	30	$-4.5155022432e-08$
6	$-2.4964571405e-01$	31	$+2.2580145600e-08$
7	$+1.6202173376e-01$	32	$-1.1290716982e-08$
8	$-9.8450913170e-02$	33	$+5.6454685428e-09$
9	$+5.7125257848e-02$	34	$-2.8227263359e-09$
10	$-3.2043727636e-02$	35	$+1.4113422414e-09$
11	$+1.7521112776e-02$	36	$-7.0565691808e-10$
12	$-9.3937359563e-03$	37	$+3.5282096728e-10$
13	$+4.9597294662e-03$	38	$-1.7640697401e-10$
14	$-2.5873265244e-03$	39	$+8.8201962240e-11$
15	$+1.3369832382e-03$	40	$-4.4100355916e-11$
16	$-6.8572831237e-04$	41	$+2.2049934531e-11$
17	$+3.4963821569e-04$	42	$-1.1024877405e-11$
18	$-1.7744974958e-04$	43	$+5.5124075668e-12$
19	$+8.9734547254e-05$	44	$-2.7561939091e-12$
20	$-4.5250271856e-05$	45	$+1.3780942688e-12$
21	$+2.2768685168e-05$	46	$-6.8904664447e-13$
22	$-1.1437476432e-05$	47	$+3.4452338541e-13$
23	$+5.7381449931e-06$	48	$-1.7226182909e-13$
24	$-2.8760584487e-06$	49	$+8.6131015376e-14$

$j = 4$**:** $\lambda_4 = -3.549616e-02$**,** $\ell_4(\mathbf{1}) = +8.740239e-02$**,** $\|\hat{v}_4 - v_4\|_2 \leq 2.61e-159$

k	$[v_4]_k$	k	$[v_4]_k$
0	$+2.9668451307e-02$	25	$+9.3012438559e-06$
1	$+2.7924054265e-01$	26	$-4.6825960255e-06$
2	$+5.6853860189e-01$	27	$+2.3534773267e-06$
3	$-4.4824188327e-01$	28	$-1.1813144791e-06$
4	$+3.3409474574e-02$	29	$+5.9235060337e-07$
5	$+2.5167800897e-01$	30	$-2.9679138185e-07$
6	$-3.4618786849e-01$	31	$+1.4861544706e-07$
7	$+3.2023977666e-01$	32	$-7.4384269422e-08$
8	$-2.4757279345e-01$	33	$+3.7218012651e-08$
9	$+1.7190363582e-01$	34	$-1.8617414047e-08$
10	$-1.1098770133e-01$	35	$+9.3112868289e-09$
11	$+6.7973144203e-02$	36	$-4.6563649096e-09$
12	$-3.9996876950e-02$	37	$+2.3283501234e-09$
13	$+2.2812107298e-02$	38	$-1.1641958726e-09$
14	$-1.2691930309e-02$	39	$+5.8208892068e-10$
15	$+6.9215101502e-03$	40	$-2.9103450369e-10$
16	$-3.7136745268e-03$	41	$+1.4551100833e-10$
17	$+1.9661656212e-03$	42	$-7.2752230647e-11$
18	$-1.0296382006e-03$	43	$+3.6374552655e-11$
19	$+5.3437088712e-04$	44	$-1.8186574860e-11$
20	$-2.7529049710e-04$	45	$+9.0929869112e-12$
21	$+1.4096381725e-04$	46	$-4.5463697302e-12$
22	$-7.1824532168e-05$	47	$+2.2731358030e-12$
23	$+3.6449132229e-05$	48	$-1.1365491915e-12$
24	$-1.8436745987e-05$	49	$+5.6826777399e-13$

$j = 5$**:** $\lambda_5 = 1.284379e-02$**,** $\ell_5(\mathbf{1}) = -3.694616e-02$**,** $\|\hat{v}_5 - v_5\|_2 \leq 1.35e-158$

k	$[v_5]_k$	k	$[v_5]_k$
0	$-8.8043043986e-03$	25	$+5.6712684176e-05$
1	$-1.2821141655e-01$	26	$-2.8984255395e-05$
2	$-4.0620172038e-01$	27	$+1.4751250734e-05$
3	$-2.5046189333e-01$	28	$-7.4813774835e-06$
4	$+5.3864569107e-01$	29	$+3.7833678490e-06$
5	$-3.4385788333e-01$	30	$-1.9087125191e-06$
6	$+1.7687881189e-02$	31	$+9.6107039237e-07$
7	$+2.1476374450e-01$	32	$-4.8314951109e-07$
8	$-3.0753683353e-01$	33	$+2.4257905525e-07$
9	$+3.0005617346e-01$	34	$-1.2166964329e-07$
10	$-2.4487939302e-01$	35	$+6.0976298735e-08$
11	$+1.7911864358e-01$	36	$-3.0539790507e-08$
12	$-1.2144537659e-01$	37	$+1.5288308267e-08$
13	$+7.7837988391e-02$	38	$-7.6505288082e-09$
14	$-4.7763889976e-02$	39	$+3.8273865267e-09$
15	$+2.8311025326e-02$	40	$-1.9143609935e-09$
16	$-1.6314781544e-02$	41	$+9.5737228056e-10$
17	$+9.1859688538e-03$	42	$-4.7873208370e-10$
18	$-5.0730711145e-03$	43	$+2.3937206451e-10$
19	$+2.7565802706e-03$	44	$-1.1968356974e-10$
20	$-1.4775041526e-03$	45	$+5.9838916142e-11$
21	$+7.8281931439e-04$	46	$-2.9917598054e-11$
22	$-4.1071277215e-04$	47	$+1.4957792419e-11$
23	$+2.1370271470e-04$	48	$-7.4783998741e-12$
24	$-1.1041605544e-04$	49	$+3.7389693930e-12$

$j = 6$**:** $\lambda_6 = -4.717778e-03$**,** $\ell_6(\mathbf{1}) = +1.534795e-02$**,** $\|\hat{v}_6 - v_6\|_2 \leq 6.52e-158$

k	$[v_6]_k$	k	$[v_6]_k$
0	$+2.9288005304e-03$	25	$+3.1381308619e-04$
1	$+5.2713511483e-02$	26	$-1.6480576749e-04$
2	$+2.5758421280e-01$	27	$+8.5893432199e-05$
3	$+3.7733209909e-01$	28	$-4.4471579252e-05$
4	$-8.7332643623e-02$	29	$+2.2894637725e-05$
5	$-3.7954659177e-01$	30	$-1.1728959324e-05$
6	$+4.8680576276e-01$	31	$+5.9836173338e-06$
7	$-2.8921038250e-01$	32	$-3.0416841901e-06$
8	$+1.4754988883e-02$	33	$+1.5415053684e-06$
9	$+1.8736691367e-01$	34	$-7.7922335306e-07$
10	$-2.7901534902e-01$	35	$+3.9304657578e-07$
11	$+2.8413322669e-01$	36	$-1.9790044778e-07$
12	$-2.4217035435e-01$	37	$+9.9495772191e-08$
13	$+1.8484043104e-01$	38	$-4.9961213509e-08$
14	$-1.3057138188e-01$	39	$+2.5062840942e-08$
15	$+8.7022433528e-02$	40	$-1.2562613434e-08$
16	$-5.5410631501e-02$	41	$+6.2929196399e-09$
17	$+3.4005412572e-02$	42	$-3.1506873855e-09$
18	$-2.0244519606e-02$	43	$+1.5768401622e-09$
19	$+1.1749626794e-02$	44	$-7.8893057108e-10$
20	$-6.6741449169e-03$	45	$+3.9463051949e-10$
21	$+3.7221436445e-03$	46	$-1.9736447204e-10$
22	$-2.0433607145e-03$	47	$+9.8694741849e-11$
23	$+1.1066109468e-03$	48	$-4.9349373601e-11$
24	$-5.9230063610e-04$	49	$+2.4674256052e-11$

$j = 7$**:** $\lambda_7 = 1.748675e-03$**,** $\ell_7(\mathbf{1}) = -6.294980e-03$**,** $\|\hat{v}_7 - v_7\|_2 \leq 3.04e-157$

k	$[v_7]_k$	k	$[v_7]_k$
0	$-9.8055287385e-04$	25	$+1.5153807365e-03$
1	$-2.1860588173e-02$	26	$-8.2791609363e-04$
2	$-1.3820176619e-01$	27	$+4.4707439112e-04$
3	$-3.4256938976e-01$	28	$-2.3895005724e-04$
4	$-1.9744018209e-01$	29	$+1.2656057412e-04$
5	$+2.9440585261e-01$	30	$-6.6499936385e-05$
6	$+8.8207678372e-02$	31	$+3.4696854350e-05$
7	$-4.1415744698e-01$	32	$-1.7991781746e-05$
8	$+4.4439275024e-01$	33	$+9.2790396749e-06$
9	$-2.5719478485e-01$	34	$-4.7629002568e-06$
10	$+1.7193970567e-02$	35	$+2.4346886257e-06$
11	$+1.6505458952e-01$	36	$-1.2400950839e-06$
12	$-2.5611455673e-01$	37	$+6.2967608130e-07$
13	$+2.7071961776e-01$	38	$-3.1887337166e-07$
14	$-2.3939660325e-01$	39	$+1.6111110867e-07$
15	$+1.8947230843e-01$	40	$-8.1243217250e-08$
16	$-1.3866088775e-01$	41	$+4.0900992244e-08$
17	$+9.5628348606e-02$	42	$-2.0562710360e-08$
18	$-6.2923782054e-02$	43	$+1.0325871095e-08$
19	$+3.9847890015e-02$	44	$-5.1803558743e-09$
20	$-2.4442242880e-02$	45	$+2.5968910133e-09$
21	$+1.4593481152e-02$	46	$-1.3009868701e-09$
22	$-8.5143110135e-03$	47	$+6.5143531766e-10$
23	$+4.8694811257e-03$	48	$-3.2605766464e-10$
24	$-2.7370962618e-03$	49	$+1.6314741925e-10$

$j = 8$**:** $\lambda_8 = -6.520209e-04$**,** $\ell_8(\mathbf{1}) = +2.557421e-03$**,** $\|\hat{v}_8 - v_8\|_2 \leq 1.39e-156$

k	$[v_8]_k$	k	$[v_8]_k$
0	$+3.3816038721e-04$	25	$+6.2014947319e-03$
1	$+8.8712621469e-03$	26	$-3.5661271091e-03$
2	$+7.0040265312e-02$	27	$+2.0174580968e-03$
3	$+2.3725221109e-01$	28	$-1.1249354472e-03$
4	$+3.2135236940e-01$	29	$+6.1924650269e-04$
5	$-4.5408229903e-02$	30	$-3.3699277531e-04$
6	$-2.9898964426e-01$	31	$+1.8152382819e-04$
7	$+1.7562043210e-01$	32	$-9.6889519810e-05$
8	$+1.8249405818e-01$	33	$+5.1294747872e-05$
9	$-4.1882715750e-01$	34	$-2.6958830190e-05$
10	$+4.1193080144e-01$	35	$+1.4076757346e-05$
11	$-2.3751440328e-01$	36	$-7.3077905994e-06$
12	$+2.2412015993e-02$	37	$+3.7742513570e-06$
13	$+1.4574225898e-01$	38	$-1.9403952820e-06$
14	$-2.3660112868e-01$	39	$+9.9356040970e-07$
15	$+2.5882669856e-01$	40	$-5.0693439770e-07$
16	$-2.3645950137e-01$	41	$+2.5784095463e-07$
17	$+1.9320544604e-01$	42	$-1.3078741619e-07$
18	$-1.4587027901e-01$	43	$+6.6183319911e-08$
19	$+1.0370377154e-01$	44	$-3.3422489643e-08$
20	$-7.0278596440e-02$	45	$+1.6848507392e-08$
21	$+4.5790881619e-02$	46	$-8.4806233509e-09$
22	$-2.8867930402e-02$	47	$+4.2632096248e-09$
23	$+1.7694940545e-02$	48	$-2.1407970004e-09$
24	$-1.0586583669e-02$	49	$+1.0740409385e-09$

$j = 9$**:** $\lambda_9 = 2.441315e-04$**,** $\ell_9(\mathbf{1}) = -1.031415e-03$**,** $\|\hat{v}_9 - v_9\|_2 \leq 6.30e-156$

k	$[v_9]_k$	k	$[v_9]_k$
0	$-1.1799547910e-04$	25	$+2.1031142393e-02$
1	$-3.5825018552e-03$	26	$-1.2881889170e-02$
2	$-3.3725281451e-02$	27	$+7.7187224649e-03$
3	$-1.4603058030e-01$	28	$-4.5360503861e-03$
4	$-2.9886824515e-01$	29	$+2.6200998185e-03$
5	$-1.8545475471e-01$	30	$-1.4902832301e-03$
6	$+2.3247141884e-01$	31	$+8.3603637339e-04$
7	$+1.3477449923e-01$	32	$-4.6322633391e-04$
8	$-2.8813177543e-01$	33	$+2.5380915511e-04$
9	$+8.1173431804e-02$	34	$-1.3767079278e-04$
10	$+2.3449479102e-01$	35	$+7.3998208039e-05$
11	$-4.1289479127e-01$	36	$-3.9448207702e-05$
12	$+3.8721655110e-01$	37	$+2.0873984602e-05$
13	$-2.2531195759e-01$	38	$-1.0971579021e-05$
14	$+2.9238457729e-02$	39	$+5.7319787919e-06$
15	$+1.2834557148e-01$	40	$-2.9783320709e-06$
16	$-2.1925984696e-01$	41	$+1.5399771601e-06$
17	$+2.4787497215e-01$	42	$-7.9277105056e-07$
18	$-2.3328565645e-01$	43	$+4.0651382364e-07$
19	$+1.9615236620e-01$	44	$-2.0772176822e-07$
20	$-1.5229855765e-01$	45	$+1.0581266317e-07$
21	$+1.1127777959e-01$	46	$-5.3752382090e-08$
22	$-7.7451963565e-02$	47	$+2.7239826076e-08$
23	$+5.1792687547e-02$	48	$-1.3774854450e-08$
24	$-3.3484691285e-02$	49	$+6.9528595371e-09$

$j = 10$**:** $\lambda_{10} = -9.168908e-05$**,** $\ell_{10}(\mathbf{1}) = +4.135796e-04$**,** $\|\hat{v}_{10} - v_{10}\|_2 \leq 2.86e-155$

k	$[v_{10}]_k$	k	$[v_{10}]_k$
0	$+4.1693739426e-05$	25	$+5.7817303214e-02$
1	$+1.4356803898e-03$	26	$-3.8259119809e-02$
2	$+1.5740329954e-02$	27	$+2.4579824204e-02$
3	$+8.2688195017e-02$	28	$-1.5389964310e-02$
4	$+2.2620743416e-01$	29	$+9.4199704682e-03$
5	$+2.7822067868e-01$	30	$-5.6509683529e-03$
6	$-1.1595749712e-02$	31	$+3.3295927307e-03$
7	$-2.7851814217e-01$	32	$-1.9304193851e-03$
8	$+8.6091010633e-02$	33	$+1.1030468741e-03$
9	$+2.1705213403e-01$	34	$-6.2204291906e-04$
10	$-2.4686942041e-01$	35	$+3.4662663351e-04$
11	$+1.2443907115e-02$	36	$-1.9106926089e-04$
12	$+2.6344547145e-01$	37	$+1.0428689691e-04$
13	$-4.0329954443e-01$	38	$-5.6410453631e-05$
14	$+3.6828131879e-01$	39	$+3.0263944566e-05$
15	$-2.1796047790e-01$	40	$-1.6115430604e-05$
16	$+3.7053787775e-02$	41	$+8.5230739066e-06$
17	$+1.1225066617e-01$	42	$-4.4797368267e-06$
18	$-2.0337663545e-01$	43	$+2.3412819003e-06$
19	$+2.3750705199e-01$	44	$-1.2173745951e-06$
20	$-2.2982635410e-01$	45	$+6.3004499121e-07$
21	$+1.9838764906e-01$	46	$-3.2470412192e-07$
22	$-1.5801420611e-01$	47	$+1.6670588380e-07$
23	$+1.1837004459e-01$	48	$-8.5295443546e-08$
24	$-8.4423852514e-02$	49	$+4.3507605710e-08$

$j = 11$**:** $\lambda_{11} = 3.451655e-05$**,** $\ell_{11}(\mathbf{1}) = -1.650674e-04$**,** $\|\hat{v}_{11} - v_{11}\|_2 \leq 1.30e-154$

k	$[v_{11}]_k$	k	$[v_{11}]_k$
0	$-1.4856799781e-05$	25	$+1.2499432422e-01$
1	$-5.7278068075e-04$	26	$-9.1176725084e-02$
2	$-7.1637694239e-03$	27	$+6.3833249597e-02$
3	$-4.4295545512e-02$	28	$-4.3160820203e-02$
4	$-1.5096183323e-01$	29	$+2.8319491938e-02$
5	$-2.7188625823e-01$	30	$-1.8099765605e-02$
6	$-1.6590624973e-01$	31	$+1.1302670227e-02$
7	$+1.8096331602e-01$	32	$-6.9136900348e-03$
8	$+1.7480530397e-01$	33	$+4.1513164595e-03$
9	$-2.3703415193e-01$	34	$-2.4513203810e-03$
10	$-2.7476522023e-02$	35	$+1.4257279586e-03$
11	$+2.4654543560e-01$	36	$-8.1788697947e-04$
12	$-2.0169921280e-01$	37	$+4.6333704418e-04$
13	$-3.6539556280e-02$	38	$-2.5948751235e-04$
14	$+2.7919354800e-01$	39	$+1.4380411081e-04$
15	$-3.9276889886e-01$	40	$-7.8929597375e-05$
16	$+3.5365056420e-01$	41	$+4.2940323581e-05$
17	$-2.1390123647e-01$	42	$-2.3171830528e-05$
18	$+4.5494057165e-02$	43	$+1.2411133618e-05$
19	$+9.7086078846e-02$	44	$-6.6021004011e-06$
20	$-1.8850600773e-01$	45	$+3.4899210672e-06$
21	$+2.2749255813e-01$	46	$-1.8341597179e-06$
22	$-2.2605014548e-01$	47	$+9.5886160127e-07$
23	$+1.9996431926e-01$	48	$-4.9884607996e-07$
24	$-1.6306691331e-01$	49	$+2.5837498578e-07$

$j = 12$**:** $\lambda_{12} = -1.301770e-05$**,** $\ell_{12}(\mathbf{1}) = +6.562912e-05$**,** $\|\hat{v}_{12} - v_{12}\|_2 \leq 5.90e-154$

k	$[v_{12}]_k$	k	$[v_{12}]_k$
0	$+5.3326453385e-06$	25	$+2.0092202477e-01$
1	$+2.2756510459e-04$	26	$-1.6749389819e-01$
2	$+3.1994533880e-03$	27	$+1.3116023367e-01$
3	$+2.2749944884e-02$	28	$-9.7694842398e-02$
4	$+9.2766484638e-02$	29	$+6.9812509799e-02$
5	$+2.1767285817e-01$	30	$-4.8161849885e-02$
6	$+2.4906057084e-01$	31	$+3.2229384441e-02$
7	$-7.2880184724e-04$	32	$-2.0999682746e-02$
8	$-2.4549995292e-01$	33	$+1.3363109696e-02$
9	$+1.0393677854e-02$	34	$-8.3258752603e-03$
10	$+2.4310591520e-01$	35	$+5.0897912214e-03$
11	$-1.6179713004e-01$	36	$-3.0584633835e-03$
12	$-1.0323446777e-01$	37	$+1.8093342105e-03$
13	$+2.4991258343e-01$	38	$-1.0552116256e-03$
14	$-1.6088907999e-01$	39	$+6.0741917329e-04$
15	$-7.1245282290e-02$	40	$-3.4548596264e-04$
16	$+2.8705639065e-01$	41	$+1.9434865776e-04$
17	$-3.8240024962e-01$	42	$-1.0822239296e-04$
18	$+3.4224992398e-01$	43	$+5.9700425540e-05$
19	$-2.1214532203e-01$	44	$-3.2649208842e-05$
20	$+5.4328744753e-02$	45	$+1.7712794772e-05$
21	$+8.2615459497e-02$	46	$-9.5385491956e-06$
22	$-1.7435718831e-01$	47	$+5.1015259849e-06$
23	$+2.1767791278e-01$	48	$-2.7112190900e-06$
24	$-2.2193728096e-01$	49	$+1.4324663259e-06$

$j = 13$**:** $\lambda_{13} = 4.916782e-06$**,** $\ell_{13}(\mathbf{1}) = -2.600972e-05$**,** $\|\hat{v}_{13} - v_{13}\|_2 \leq 2.69e-153$

k	$[v_{13}]_k$	k	$[v_{13}]_k$
0	$-1.9251350393e-06$	25	$+2.0795837756e-01$
1	$-9.0120688193e-05$	26	$-2.1747612414e-01$
2	$-1.4071545989e-03$	27	$+2.0129170113e-01$
3	$-1.1313434202e-02$	28	$-1.7132377568e-01$
4	$-5.3659908959e-02$	29	$+1.3687437652e-01$
5	$-1.5458581942e-01$	30	$-1.0396376707e-01$
6	$-2.5163194528e-01$	31	$+7.5729727003e-02$
7	$-1.5007440271e-01$	32	$-5.3236245942e-02$
8	$+1.5241951743e-01$	33	$+3.6289385302e-02$
9	$+1.8094865039e-01$	34	$-2.4077661904e-02$
10	$-1.7192716230e-01$	35	$+1.5596595759e-02$
11	$-1.1489889704e-01$	36	$-9.8881452546e-03$
12	$+2.4557237593e-01$	37	$+6.1487178908e-03$
13	$-8.8591648220e-02$	38	$-3.7568145553e-03$
14	$-1.5055178898e-01$	39	$+2.2588886915e-03$
15	$+2.4112491343e-01$	40	$-1.3384391950e-03$
16	$-1.2648307898e-01$	41	$+7.8243844468e-04$
17	$-9.5709446764e-02$	42	$-4.5176295674e-04$
18	$+2.9004328791e-01$	43	$+2.5786523736e-04$
19	$-3.7260755967e-01$	44	$-1.4563600466e-04$
20	$+3.3329298229e-01$	45	$+8.1447123843e-05$
21	$-2.1203246136e-01$	46	$-4.5135807056e-05$
22	$+6.3402917487e-02$	47	$+2.4802003361e-05$
23	$+6.8682907158e-02$	48	$-1.3521708824e-05$
24	$-1.6073398510e-01$	49	$+7.3180468444e-06$

$j = 14$**:** $\lambda_{14} = -1.859307e-06$**,** $\ell_{14}(\mathbf{1}) = +1.027988e-05$**,** $\|\hat{v}_{14} - v_{14}\|_2 \leq 1.23e-152$

k	$[v_{14}]_k$	k	$[v_{14}]_k$
0	$+6.9839204447e-07$	25	$+5.5183102296e-02$
1	$+3.5590437909e-05$	26	$-1.4750093339e-01$
2	$+6.1120610499e-04$	27	$+1.9826164119e-01$
3	$+5.4824014243e-03$	28	$-2.1266086666e-01$
4	$+2.9643512430e-02$	29	$+2.0109845198e-01$
5	$+1.0100817539e-01$	30	$-1.7457909549e-01$
6	$+2.1089546142e-01$	31	$+1.4214115435e-01$
7	$+2.2613606565e-01$	32	$-1.0997004731e-01$
8	$+3.8586220423e-03$	33	$+8.1561628605e-02$
9	$-2.2222524508e-01$	34	$-5.8359654245e-02$
10	$-2.8641204973e-02$	35	$+4.0479930522e-02$
11	$+2.2532854208e-01$	36	$-2.7321283018e-02$
12	$-6.5066300773e-02$	37	$+1.7997574694e-02$
13	$-1.8658213373e-01$	38	$-1.1600170958e-02$
14	$+2.1834492290e-01$	39	$+7.3310127987e-03$
15	$-2.7552311101e-02$	40	$-4.5508207970e-03$
16	$-1.7861462258e-01$	41	$+2.7791521272e-03$
17	$+2.2733026157e-01$	42	$-1.6719256136e-03$
18	$-9.8399813265e-02$	43	$+9.9201702611e-04$
19	$-1.1276964385e-01$	44	$-5.8113457693e-04$
20	$+2.8992422252e-01$	45	$+3.3643364330e-04$
21	$-3.6350594136e-01$	46	$-1.9264434256e-04$
22	$+3.2619624408e-01$	47	$+1.0918968937e-04$
23	$-2.1310346976e-01$	48	$-6.1302784273e-05$
24	$+7.2607085163e-02$	49	$+3.4113866789e-05$

$j = 15$**:** $\lambda_{15} = 7.038113e-07$**,** $\ell_{15}(\mathbf{1}) = -4.053403e-06$**,** $\|\hat{v}_{15} - v_{15}\|_2 \leq 5.67e-152$

k	$[v_{15}]_k$	k	$[v_{15}]_k$
0	$-2.5440017920e-07$	25	$-2.1502807213e-01$
1	$-1.4022477968e-05$	26	$+8.1860360750e-02$
2	$-2.6272062088e-04$	27	$+4.2043984978e-02$
3	$-2.6012069807e-03$	28	$-1.3456317163e-01$
4	$-1.5788943361e-02$	29	$+1.8853772795e-01$
5	$-6.2043814076e-02$	30	$-2.0749004043e-01$
6	$-1.5722370765e-01$	31	$+2.0036342236e-01$
7	$-2.3565390879e-01$	32	$-1.7727808525e-01$
8	$-1.3561427070e-01$	33	$+1.4696337733e-01$
9	$+1.3325448741e-01$	34	$-1.1570102844e-01$
10	$+1.8086453250e-01$	35	$+8.7286618375e-02$
11	$-1.2714887956e-01$	36	$-6.3509047340e-02$
12	$-1.5026485342e-01$	37	$+4.4781927369e-02$
13	$+1.9297368195e-01$	38	$-3.0717812120e-02$
14	$+2.9309212784e-02$	39	$+2.0559725140e-02$
15	$-2.1910446217e-01$	40	$-1.3460745442e-02$
16	$+1.8071658077e-01$	41	$+8.6388384265e-03$
17	$+2.0239796958e-02$	42	$-5.4444040770e-03$
18	$-1.9415530102e-01$	43	$+3.3745964472e-03$
19	$+2.1214228010e-01$	44	$-2.0599280735e-03$
20	$-7.5900765837e-02$	45	$+1.2398053912e-03$
21	$-1.2440590241e-01$	46	$-7.3651436339e-04$
22	$+2.8777318166e-01$	47	$+4.3225702156e-04$
23	$-3.5507923797e-01$	48	$-2.5084320874e-04$
24	$+3.2051989679e-01$	49	$+1.4404356517e-04$

$j = 16$**:** $\lambda_{16} = -2.666413e-07$**,** $\ell_{16}(\mathbf{1}) = +1.595012e-06$**,** $\|\hat{v}_{16} - v_{16}\|_2 \leq 2.61e-151$

k	$[v_{16}]_k$	k	$[v_{16}]_k$
0	$+9.2996230395e-08$	25	$-3.4725646475e-01$
1	$+5.5135688933e-06$	26	$+3.1592718589e-01$
2	$+1.1193313080e-04$	27	$-2.1756182882e-01$
3	$+1.2126129608e-03$	28	$+9.1100549291e-02$
4	$+8.1636033175e-03$	29	$+2.9216186478e-02$
5	$+3.6316728617e-02$	30	$-1.2185392133e-01$
6	$+1.0792745979e-01$	31	$+1.7875253842e-01$
7	$+2.0505505174e-01$	32	$-2.0196552123e-01$
8	$+2.0743349642e-01$	33	$+1.9910506297e-01$
9	$+4.1733275532e-03$	34	$-1.7943588558e-01$
10	$-2.0334560247e-01$	35	$+1.5134273703e-01$
11	$-5.3398498140e-02$	36	$-1.2114474722e-01$
12	$+2.0454995644e-01$	37	$+9.2884488580e-02$
13	$-2.7238319182e-03$	38	$-6.8662512632e-02$
14	$-2.0186731146e-01$	39	$+4.9176688427e-02$
15	$+1.3107653491e-01$	40	$-3.4254239815e-02$
16	$+9.9945226519e-02$	41	$+2.3276032422e-02$
17	$-2.2698787815e-01$	42	$-1.5467844597e-02$
18	$+1.4199152968e-01$	43	$+1.0073630785e-02$
19	$+5.6508088239e-02$	44	$-6.4409576907e-03$
20	$-2.0172377992e-01$	45	$+4.0493818697e-03$
21	$+1.9734367649e-01$	46	$-2.5065755768e-03$
22	$-5.8123606009e-02$	47	$+1.5294552637e-03$
23	$-1.3200862239e-01$	48	$-9.2089308034e-04$
24	$+2.8425834978e-01$	49	$+5.4764936320e-04$

$j = 17$**:** $\lambda_{17} = 1.010906e-07$**,** $\ell_{17}(\mathbf{1}) = -6.265117e-07$**,** $\|\hat{v}_{17} - v_{17}\|_2 \leq 1.21e-150$

k	$[v_{17}]_k$	k	$[v_{17}]_k$
0	$-3.4098551032e-08$	25	$-1.3656541888e-01$
1	$-2.1640736846e-06$	26	$+2.7980383065e-01$
2	$-4.7328370576e-05$	27	$-3.3994753162e-01$
3	$-5.5690960434e-04$	28	$+3.1215642565e-01$
4	$-4.1180222676e-03$	29	$-2.2051930465e-01$
5	$-2.0448659427e-02$	30	$+1.0027807240e-01$
6	$-6.9606992142e-02$	31	$+1.6666296356e-02$
7	$-1.5914504313e-01$	32	$-1.0932638994e-01$
8	$-2.2227375851e-01$	33	$+1.6888356888e-01$
9	$-1.2260963819e-01$	34	$-1.9609184136e-01$
10	$+1.2076214445e-01$	35	$+1.9734000776e-01$
11	$+1.7649142663e-01$	36	$-1.8106544613e-01$
12	$-9.2684322852e-02$	37	$+1.5528017812e-01$
13	$-1.6719411743e-01$	38	$-1.2628987850e-01$
14	$+1.4401093290e-01$	39	$+9.8336217071e-02$
15	$+9.5602693572e-02$	40	$-7.3799094751e-02$
16	$-2.0484006442e-01$	41	$+5.3645880739e-02$
17	$+6.4731864945e-02$	42	$-3.7917312144e-02$
18	$+1.4809281929e-01$	43	$+2.6138849977e-02$
19	$-2.2068691126e-01$	44	$-1.7618679083e-02$
20	$+1.0624231190e-01$	45	$+1.1636124136e-02$
21	$+8.3532790173e-02$	46	$-7.5433439992e-03$
22	$-2.0430717426e-01$	47	$+4.8073369108e-03$
23	$+1.8376263311e-01$	48	$-3.0158419719e-03$
24	$-4.4267212927e-02$	49	$+1.8645938566e-03$

$j = 18$**:** $\lambda_{18} = -3.834970e-08$**,** $\ell_{18}(\mathbf{1}) = +2.457002e-07$**,** $\|\hat{v}_{18} - v_{18}\|_2 \leq 5.59e-150$

k	$[v_{18}]_k$	k	$[v_{18}]_k$
0	$+1.2536120551e-08$	25	$+1.7172887351e-01$
1	$+8.4807341810e-07$	26	$-3.3644388194e-02$
2	$+1.9880091518e-05$	27	$-1.3878722065e-01$
3	$+2.5250997382e-04$	28	$+2.7468309857e-01$
4	$+2.0343801499e-03$	29	$-3.3305994586e-01$
5	$+1.1151032091e-02$	30	$+3.0900148747e-01$
6	$+4.2734393241e-02$	31	$-2.2375675350e-01$
7	$+1.1381885071e-01$	32	$+1.0935212066e-01$
8	$+1.9979660368e-01$	33	$+4.3724105691e-03$
9	$+1.9145447805e-01$	34	$-9.6948356775e-02$
10	$+2.3229228209e-03$	35	$+1.5891698730e-01$
11	$-1.8869298308e-01$	36	$-1.8987570196e-01$
12	$-6.7934583203e-02$	37	$+1.9508369697e-01$
13	$+1.8225094020e-01$	38	$-1.8217818845e-01$
14	$+4.1748510917e-02$	39	$+1.5877619403e-01$
15	$-1.9608195552e-01$	40	$-1.3112571452e-01$
16	$+5.8204060839e-02$	41	$+1.0362382477e-01$
17	$+1.5681113453e-01$	42	$-7.8898679318e-02$
18	$-1.8049254621e-01$	43	$+5.8171488577e-02$
19	$+4.9198206331e-03$	44	$-4.1693557471e-02$
20	$+1.7841017445e-01$	45	$+2.9139951766e-02$
21	$-2.0695023006e-01$	46	$-1.9909739735e-02$
22	$+7.4944967825e-02$	47	$+1.3326373626e-02$
23	$+1.0340618680e-01$	48	$-8.7538933653e-03$
24	$-2.0384448613e-01$	49	$+5.6519408554e-03$

$j = 19$**:** $\lambda_{19} = 1.455614e-08$**,** $\ell_{19}(\mathbf{1}) = -9.622052e-08$**,** $\|\hat{v}_{19} - v_{19}\|_2 \leq 2.59e-149$

k	$[v_{19}]_k$	k	$[v_{19}]_k$
0	$-4.6196522899e-09$	25	$+1.1784115736e-01$
1	$-3.3188936729e-07$	26	$-2.0158750590e-01$
2	$-8.3023730111e-06$	27	$+1.6131431421e-01$
3	$-1.1322378885e-04$	28	$-2.5685267346e-02$
4	$-9.8719943622e-04$	29	$-1.3919293855e-01$
5	$-5.9192441604e-03$	30	$+2.6907469761e-01$
6	$-2.5203893273e-02$	31	$-3.2650636071e-01$
7	$-7.6474071929e-02$	32	$+3.0629800540e-01$
8	$-1.6048505755e-01$	33	$-2.2716060579e-01$
9	$-2.1063495521e-01$	34	$+1.1828815140e-01$
10	$-1.1067969063e-01$	35	$-7.6789087845e-03$
11	$+1.1239356380e-01$	36	$-8.4698446206e-02$
12	$+1.7086512800e-01$	37	$+1.4884558178e-01$
13	$-6.7538992861e-02$	38	$-1.8332561717e-01$
14	$-1.7165461730e-01$	39	$+1.9235082731e-01$
15	$+9.9035014500e-02$	40	$-1.8278450402e-01$
16	$+1.3533774098e-01$	41	$+1.6183106417e-01$
17	$-1.6981569664e-01$	42	$-1.3564216425e-01$
18	$-2.2210476777e-02$	43	$+1.0873027636e-01$
19	$+1.8617134569e-01$	44	$-8.3941908372e-02$
20	$-1.4361631722e-01$	45	$+6.2735787968e-02$
21	$-4.4733303701e-02$	46	$-4.5569311557e-02$
22	$+1.9568504377e-01$	47	$+3.2270578369e-02$
23	$-1.8990383793e-01$	48	$-2.2336838078e-02$
24	$+4.8345292430e-02$	49	$+1.5143776347e-02$

$j = 20$**:** $\lambda_{20} = -5.527568e-09$**,** $\ell_{20}(\mathbf{1}) = +3.763392e-08$**,** $\|\hat{v}_{20} - v_{20}\|_2 \leq 1.20e-148$

k	$[v_{20}]_k$	k	$[v_{20}]_k$
0	$+1.7059246344e-09$	25	$-1.7197266599e-01$
1	$+1.2972248512e-07$	26	$+2.6147097823e-02$
2	$+3.4495679690e-06$	27	$+1.2817274074e-01$
3	$+5.0275713287e-05$	28	$-1.9834031291e-01$
4	$+4.7166922523e-04$	29	$+1.5246147155e-01$
5	$+3.0705964380e-03$	30	$-1.9924058437e-02$
6	$+1.4375423448e-02$	31	$-1.3816663689e-01$
7	$+4.8885722595e-02$	32	$+2.6309633388e-01$
8	$+1.1887846966e-01$	33	$-3.2020764939e-01$
9	$+1.9488399384e-01$	34	$+3.0391348094e-01$
10	$+1.7739173543e-01$	35	$-2.3063961639e-01$
11	$-8.4328050351e-04$	36	$+1.2705622557e-01$
12	$-1.7706713617e-01$	37	$-1.9494769483e-02$
13	$-7.6550582688e-02$	38	$-7.2563495516e-02$
14	$+1.6230633866e-01$	39	$+1.3866728575e-01$
15	$+7.1039066088e-02$	40	$-1.7645164731e-01$
16	$-1.7760845889e-01$	41	$+1.8915568161e-01$
17	$-2.1134169777e-03$	42	$-1.8289413340e-01$
18	$+1.7868088205e-01$	43	$+1.6444504563e-01$
19	$-1.1743579090e-01$	44	$-1.3982976486e-01$
20	$-8.5648360077e-02$	45	$+1.1363941219e-01$
21	$+1.9261939711e-01$	46	$-8.8910120073e-02$
22	$-1.0310831877e-01$	47	$+6.7321331191e-02$
23	$-8.4011964101e-02$	48	$-4.9530742098e-02$
24	$+2.0386780236e-01$	49	$+3.5521478683e-02$

$j = 21$**:** $\lambda_{21} = 2.099914e-09$**,** $\ell_{21}(\mathbf{1}) = -1.470261e-08$**,** $\|\hat{v}_{21} - v_{21}\|_2 \leq 5.60e-148$

k	$[v_{21}]_k$	k	$[v_{21}]_k$
0	$-6.3112920693e-10$	25	$-1.1407814914e-01$
1	$-5.0646933860e-08$	26	$+2.0594464494e-01$
2	$-1.4267534045e-06$	27	$-1.5452204276e-01$
3	$-2.2132562200e-05$	28	$+7.8511071689e-03$
4	$-2.2231306122e-04$	29	$+1.3541707196e-01$
5	$-1.5614631328e-03$	30	$-1.9461525185e-01$
6	$-7.9695262978e-03$	31	$+1.4505234478e-01$
7	$-2.9996517155e-02$	32	$-1.5981870119e-02$
8	$-8.2736317963e-02$	33	$-1.3599652870e-01$
9	$-1.6133355966e-01$	34	$+2.5682622485e-01$
10	$-2.0019411599e-01$	35	$-3.1409379431e-01$
11	$-9.9623682490e-02$	36	$+3.0174008355e-01$
12	$+1.0684287787e-01$	37	$-2.3411938955e-01$
13	$+1.6482106769e-01$	38	$+1.3562988243e-01$
14	$-4.8949229275e-02$	39	$-3.1077488275e-02$
15	$-1.7022749919e-01$	40	$-6.0536652640e-02$
16	$+6.2276720019e-02$	41	$+1.2838409429e-01$
17	$+1.5402826841e-01$	42	$-1.6926519333e-01$
18	$-1.2505999281e-01$	43	$+1.8551237253e-01$
19	$-8.5550117104e-02$	44	$-1.8251645775e-01$
20	$+1.8264489364e-01$	45	$+1.6661852901e-01$
21	$-5.8125032449e-02$	46	$-1.4367970005e-01$
22	$-1.3025578142e-01$	47	$+1.1833590273e-01$
23	$+1.8425252641e-01$	48	$-9.3785307087e-02$
24	$-6.3808192716e-02$	49	$+7.1910939578e-02$

$j = 22$**:** $\lambda_{22} = -7.980458e-10$**,** $\ell_{22}(\mathbf{1}) = +5.737986e-09$**,** $\|\hat{v}_{22} - v_{22}\|_2 \leq 2.61e-147$

k	$[v_{22}]_k$	k	$[v_{22}]_k$
0	$+2.3388513408e-10$	25	$+1.6720760999e-01$
1	$+1.9753910029e-08$	26	$-2.8053360802e-02$
2	$+5.8770580534e-07$	27	$-1.3649800760e-01$
3	$+9.6686981119e-06$	28	$+2.0407453226e-01$
4	$+1.0353185802e-04$	29	$-1.3827283917e-01$
5	$+7.8033021831e-04$	30	$-7.0836196281e-03$
6	$+4.3113729372e-03$	31	$+1.4033841488e-01$
7	$+1.7783559924e-02$	32	$-1.9073408544e-01$
8	$+5.4770239462e-02$	33	$+1.3894524396e-01$
9	$+1.2323822761e-01$	34	$-1.3550371867e-02$
10	$+1.9017167299e-01$	35	$-1.3290188898e-01$
11	$+1.6471447265e-01$	36	$+2.5031673431e-01$
12	$-4.7850796728e-03$	37	$-3.0810373645e-01$
13	$-1.6776969302e-01$	38	$+2.9968934129e-01$
14	$-8.1296996863e-02$	39	$-2.3753848648e-01$
15	$+1.4506886517e-01$	40	$+1.4398536699e-01$
16	$+9.0374064261e-02$	41	$-4.2425725335e-02$
17	$-1.5559774285e-01$	42	$-4.8615972325e-02$
18	$-4.6243310937e-02$	43	$+1.1800125112e-01$
19	$+1.7507430693e-01$	44	$-1.6177883397e-01$
20	$-5.1351426055e-02$	45	$+1.8143502367e-01$
21	$-1.4042756998e-01$	46	$-1.8166072495e-01$
22	$+1.6152275788e-01$	47	$+1.6835216575e-01$
23	$-2.3301810388e-03$	48	$-1.4718382205e-01$
24	$-1.5825917493e-01$	49	$+1.2280521956e-01$

$j = 23$: $\lambda_{23} = 3.033863e-10$, $\ell_{23}(\mathbf{1}) = -2.237249e-09$, $\|\hat{v}_{23} - v_{23}\|_2 \leq 1.22e-146$

k	$[v_{23}]_k$	k	$[v_{23}]_k$
0	$-8.6804571889e-11$	25	$+4.5285570024e-02$
1	$-7.6975883194e-09$	26	$-1.7310729218e-01$
2	$-2.4119731656e-07$	27	$+1.4574336073e-01$
3	$-4.1948155614e-06$	28	$+3.2620016822e-03$
4	$-4.7702008988e-05$	29	$-1.5281166826e-01$
5	$-3.8401774639e-04$	30	$+1.9977731335e-01$
6	$-2.2831551706e-03$	31	$-1.2356150571e-01$
7	$-1.0237480310e-02$	32	$-1.9167514366e-02$
8	$-3.4786144530e-02$	33	$+1.4350821274e-01$
9	$-8.8461997270e-02$	34	$-1.8689373833e-01$
10	$-1.6174907672e-01$	35	$+1.3399415336e-01$
11	$-1.9060592490e-01$	36	$-1.2377780355e-02$
12	$-8.9273138066e-02$	37	$-1.2905185766e-01$
13	$+1.0322918430e-01$	38	$+2.4360322791e-01$
14	$+1.5884384712e-01$	39	$-3.0218476337e-01$
15	$-3.5291651870e-02$	40	$+2.9768817511e-01$
16	$-1.6572870469e-01$	41	$-2.4084561045e-01$
17	$+3.2815011623e-02$	42	$+1.5210109376e-01$
18	$+1.6032218229e-01$	43	$-5.3535337935e-02$
19	$-8.2146566674e-02$	44	$-3.6803359985e-02$
20	$-1.2357119205e-01$	45	$+1.0752662689e-01$
21	$+1.5208919982e-01$	46	$-1.5400619319e-01$
22	$+2.0318168609e-02$	47	$+1.7693790421e-01$
23	$-1.6811880553e-01$	48	$-1.8033622563e-01$
24	$+1.2713595975e-01$	49	$+1.6964697237e-01$

$j = 24$: $\lambda_{24} = -1.153695e-10$, $\ell_{24}(\mathbf{1}) = +8.715538e-10$, $\|\hat{v}_{24} - v_{24}\|_2 \leq 5.68e-146$

k	$[v_{24}]_k$	k	$[v_{24}]_k$
0	$+3.2260982235e-11$	25	$-1.7443291719e-01$
1	$+2.9970391679e-09$	26	$+8.7694907819e-02$
2	$+9.8658898015e-08$	27	$+8.3388961894e-02$
3	$+1.8086751407e-06$	28	$-1.7820267686e-01$
4	$+2.1768814091e-05$	29	$+1.2264724459e-01$
5	$+1.8641797134e-04$	30	$+3.0027512598e-02$
6	$+1.1865918093e-03$	31	$-1.6436808299e-01$
7	$+5.7450758300e-03$	32	$+1.9410290130e-01$
8	$+2.1333514019e-02$	33	$-1.1050022868e-01$
9	$+6.0390944436e-02$	34	$-2.8852558302e-02$
10	$+1.2699126303e-01$	35	$+1.4535253454e-01$
11	$+1.8556184101e-01$	36	$-1.8320924090e-01$
12	$+1.5307489812e-01$	37	$+1.3005851741e-01$
13	$-9.1921527600e-03$	38	$-1.2257384996e-02$
14	$-1.6023014082e-01$	39	$-1.2457873557e-01$
15	$-8.3530590452e-02$	40	$+2.3670990710e-01$
16	$+1.3059964651e-01$	41	$-2.9629172774e-01$
17	$+1.0275567550e-01$	42	$+2.9567590109e-01$
18	$-1.3337106208e-01$	43	$-2.4399753960e-01$
19	$-7.7500253821e-02$	44	$+1.5995727127e-01$
20	$+1.5884275331e-01$	45	$-6.4400035097e-02$
21	$+4.3982794937e-03$	46	$-2.5103762820e-02$
22	$-1.6115909395e-01$	47	$+9.6970234771e-02$
23	$+1.0550307284e-01$	48	$-1.4596182935e-01$
24	$+7.8951045167e-02$	49	$+1.7203552842e-01$

$j = 25$**:** $\lambda_{25} = 4.388345e-11$, $\quad \ell_{25}(\mathbf{1}) = -3.392576e-10$, $\quad \|\hat{v}_{25} - v_{25}\|_2 \leq 2.65e-145$

k	$[v_{25}]_k$	k	$[v_{25}]_k$
0	$-1.2004798857e-11$	25	$+5.0868194365e-02$
1	$-1.1659941593e-09$	26	$+1.2124394340e-01$
2	$-4.0232865913e-08$	27	$-1.6579902758e-01$
3	$-7.7546769681e-07$	28	$+4.8300081486e-02$
4	$-9.8486812230e-06$	29	$+1.1235821092e-01$
5	$-8.9393771905e-05$	30	$-1.7643370655e-01$
6	$-6.0649683811e-04$	31	$+9.9652310905e-02$
7	$-3.1527747726e-03$	32	$+5.2526145634e-02$
8	$-1.2695420231e-02$	33	$-1.7228288246e-01$
9	$-3.9543231573e-02$	34	$+1.8776473276e-01$
10	$-9.3702479929e-02$	35	$-9.9074515311e-02$
11	$-1.6177283367e-01$	36	$-3.6525641803e-02$
12	$-1.8163749309e-01$	37	$+1.4618863292e-01$
13	$-7.9506782170e-02$	38	$-1.7974194723e-01$
14	$+1.0098240660e-01$	39	$+1.2700777520e-01$
15	$+1.5310714279e-01$	40	$-1.3018363122e-02$
16	$-2.5289911146e-02$	41	$-1.1958749498e-01$
17	$-1.5991364349e-01$	42	$+2.2965369797e-01$
18	$+9.7098207038e-03$	43	$-2.9038624057e-01$
19	$+1.5898323358e-01$	44	$+2.9360193875e-01$
20	$-4.4222390707e-02$	45	$-2.4695758792e-01$
21	$-1.4326661134e-01$	46	$+1.6753563800e-01$
22	$+1.1199762852e-01$	47	$-7.5011888517e-02$
23	$+7.7666370784e-02$	48	$-1.3524539790e-02$
24	$-1.6509458703e-01$	49	$+8.6343846240e-02$

$j = 26$**:** $\lambda_{26} = -1.669605e-11$, $\quad \ell_{26}(\mathbf{1}) = +1.319616e-10$, $\quad \|\hat{v}_{26} - v_{26}\|_2 \leq 1.24e-144$

k	$[v_{26}]_k$	k	$[v_{26}]_k$
0	$+4.4722743877e-12$	25	$+1.2747585372e-01$
1	$+4.5330759883e-10$	26	$-1.4575267712e-01$
2	$+1.6361363208e-08$	27	$-2.0742960210e-03$
3	$+3.3078118645e-07$	28	$+1.4793597636e-01$
4	$+4.4209759051e-06$	29	$-1.4774328350e-01$
5	$+4.2397026300e-05$	30	$+1.1796927025e-02$
6	$+3.0540915187e-04$	31	$+1.3332640115e-01$
7	$+1.6962950487e-03$	32	$-1.7007454931e-01$
8	$+7.3594297334e-03$	33	$+7.7773350829e-02$
9	$+2.4991321128e-02$	34	$+7.1211774770e-02$
10	$+6.5751653162e-02$	35	$-1.7744911280e-01$
11	$+1.3020522049e-01$	36	$+1.8123906368e-01$
12	$+1.8098791979e-01$	37	$-8.9202083580e-02$
13	$+1.4223300434e-01$	38	$-4.2511608235e-02$
14	$-1.3866208559e-02$	39	$+1.4625253937e-01$
15	$-1.5403304773e-01$	40	$-1.7651818615e-01$
16	$-8.4071197871e-02$	41	$+1.2472304338e-01$
17	$+1.1860830536e-01$	42	$-1.4518508832e-02$
18	$+1.1043738934e-01$	43	$-1.1416265678e-01$
19	$-1.1275818123e-01$	44	$+2.2244686771e-01$
20	$-9.8588406146e-02$	45	$-2.8443590727e-01$
21	$+1.3661131209e-01$	46	$+2.9142404039e-01$
22	$+4.7921352957e-02$	47	$-2.4969444505e-01$
23	$-1.5978542906e-01$	48	$+1.7481927615e-01$
24	$+4.6669315777e-02$	49	$-8.5361737846e-02$

$j = 27$**:** $\lambda_{27} = 6.353612e-12, \quad \ell_{27}(\mathbf{1}) = -5.129468e-11, \quad \|\hat{v}_{27} - v_{27}\|_2 \leq 5.81e-144$

k	$[v_{27}]_k$	k	$[v_{27}]_k$
0	$-1.6678529109e-12$	25	$-1.3914086561e-01$
1	$-1.7611878204e-10$	26	$-1.8360931012e-02$
2	$-6.6366776908e-09$	27	$+1.5326777694e-01$
3	$-1.4043719272e-07$	28	$-1.1293858917e-01$
4	$-1.9704205009e-06$	29	$-4.8190248616e-02$
5	$-1.9907758705e-05$	30	$+1.6149437371e-01$
6	$-1.5174299525e-04$	31	$-1.2448872146e-01$
7	$-8.9670145640e-04$	32	$-2.0458946920e-02$
8	$-4.1688071647e-03$	33	$+1.4765874501e-01$
9	$-1.5318812094e-02$	34	$-1.6083560105e-01$
10	$-4.4245056897e-02$	35	$+5.7552969245e-02$
11	$-9.8496969381e-02$	36	$+8.6587421919e-02$
12	$-1.6143502938e-01$	37	$-1.8056731831e-01$
13	$-1.7312769050e-01$	38	$+1.7483665549e-01$
14	$-7.0233012670e-02$	39	$-8.0767975887e-02$
15	$+9.9709589379e-02$	40	$-4.7080167374e-02$
16	$+1.4767149466e-01$	41	$+1.4571968937e-01$
17	$-1.8046111286e-02$	42	$-1.7354165307e-01$
18	$-1.5369603823e-01$	43	$+1.2309728750e-01$
19	$-8.2515820559e-03$	44	$-1.6638504288e-02$
20	$+1.5338811466e-01$	45	$-1.0837331381e-01$
21	$-1.2525620684e-02$	46	$+2.1509879697e-01$
22	$-1.4986993665e-01$	47	$-2.7841363423e-01$
23	$+7.0553094429e-02$	48	$+2.8910690930e-01$
24	$+1.1467959219e-01$	49	$-2.5218129152e-01$

$j = 28$**:** $\lambda_{28} = -2.418317e-12, \quad \ell_{28}(\mathbf{1}) = +1.992622e-11, \quad \|\hat{v}_{28} - v_{28}\|_2 \leq 2.72e-143$

k	$[v_{28}]_k$	k	$[v_{28}]_k$
0	$+6.2259699081e-13$	25	$-6.0018035421e-03$
1	$+6.8384105752e-11$	26	$+1.4821629920e-01$
2	$+2.6857270091e-09$	27	$-9.6886382951e-02$
3	$+5.9368284938e-08$	28	$-7.2821118323e-02$
4	$+8.7249916911e-07$	29	$+1.5894187035e-01$
5	$+9.2631112781e-06$	30	$-7.4222462427e-02$
6	$+7.4483590687e-05$	31	$-8.5429757336e-02$
7	$+4.6656415714e-04$	32	$+1.6483633940e-01$
8	$+2.3134779413e-03$	33	$-9.9029974134e-02$
9	$+9.1420496539e-03$	34	$-4.8030767726e-02$
10	$+2.8728725901e-02$	35	$+1.5668966930e-01$
11	$+7.0855855226e-02$	36	$-1.4996216517e-01$
12	$+1.3293070277e-01$	37	$+3.9232777349e-02$
13	$+1.7640354496e-01$	38	$+9.9142229485e-02$
14	$+1.3201943807e-01$	39	$-1.8217965487e-01$
15	$-1.8677965771e-02$	40	$+1.6875367829e-01$
16	$-1.4886307819e-01$	41	$-7.3645122644e-02$
17	$-8.3449668866e-02$	42	$-5.0453843641e-02$
18	$+1.0876293703e-01$	43	$+1.4472027591e-01$
19	$+1.1491455318e-01$	44	$-1.7080167652e-01$
20	$-9.4405488010e-02$	45	$+1.2203473219e-01$
21	$-1.1218513138e-01$	46	$-1.9277411649e-02$
22	$+1.1271701541e-01$	47	$-1.0227683431e-01$
23	$+7.9512833976e-02$	48	$+2.0761718511e-01$
24	$-1.4465920757e-01$	49	$-2.7229701434e-01$

$j = 29$**:** $\lambda_{29} = 9.206274e-13$**,** $\ell_{29}(\mathbf{1}) = -7.736140e-12$**,** $\|\hat{v}_{29} - v_{29}\|_2 \leq 1.28e-142$

k	$[v_{29}]_k$	k	$[v_{29}]_k$
0	$-2.3261861851e-13$	25	$+1.3427571238e-01$
1	$-2.6537502006e-11$	26	$-1.0197852170e-01$
2	$-1.0845014304e-09$	27	$-7.1978755758e-02$
3	$-2.4997977908e-08$	28	$+1.5206988491e-01$
4	$-3.8403281045e-07$	29	$-4.6136747755e-02$
5	$-4.2744181205e-06$	30	$-1.1268847280e-01$
6	$-3.6158761725e-05$	31	$+1.4973968839e-01$
7	$-2.3930616479e-04$	32	$-3.4776951085e-02$
8	$-1.2604759035e-03$	33	$-1.1358212855e-01$
9	$-5.3283302399e-03$	34	$+1.6070737855e-01$
10	$-1.8085705768e-02$	35	$-7.3365326757e-02$
11	$-4.8873347785e-02$	36	$-7.1010771848e-02$
12	$-1.0287576572e-01$	37	$+1.6160646294e-01$
13	$-1.6075886746e-01$	38	$-1.3833695416e-01$
14	$-1.6496213167e-01$	39	$+2.2868808827e-02$
15	$-6.1383129450e-02$	40	$+1.0932184791e-01$
16	$+9.9134249551e-02$	41	$-1.8270160139e-01$
17	$+1.4253640965e-01$	42	$+1.6310751503e-01$
18	$-1.2902955864e-02$	43	$-6.7706038275e-02$
19	$-1.4757770894e-01$	44	$-5.2815649345e-02$
20	$-2.2110477079e-02$	45	$+1.4335067125e-01$
21	$+1.4559570653e-01$	46	$-1.6827874672e-01$
22	$+1.3184876153e-02$	47	$+1.2144992745e-01$
23	$-1.4796068625e-01$	48	$-2.2349122311e-02$
24	$+3.2506603709e-02$	49	$-9.5921615504e-02$

$j = 30$**:** $\lambda_{30} = -3.505309e-13$**,** $\ell_{30}(\mathbf{1}) = +3.001843e-12$**,** $\|\hat{v}_{30} - v_{30}\|_2 \leq 5.98e-142$

k	$[v_{30}]_k$	k	$[v_{30}]_k$
0	$+8.6984479598e-14$	25	$-1.2243155723e-01$
1	$+1.0292879842e-11$	26	$-4.8273218736e-02$
2	$+4.3704308320e-10$	27	$+1.4777487381e-01$
3	$+1.0487295291e-08$	28	$-4.2716704650e-02$
4	$+1.6810211529e-07$	29	$-1.1789356976e-01$
5	$+1.9573959706e-06$	30	$+1.3417651497e-01$
6	$+1.7377221324e-05$	31	$+4.0423095824e-03$
7	$+1.2115557786e-04$	32	$-1.3775157911e-01$
8	$+6.7546704760e-04$	33	$+1.3067010487e-01$
9	$+3.0407254436e-03$	34	$+2.2192866315e-03$
10	$+1.1080686297e-02$	35	$-1.3341034753e-01$
11	$+3.2521637753e-02$	36	$+1.5143223821e-01$
12	$+7.5706318333e-02$	37	$-4.8746625822e-02$
13	$+1.3520655160e-01$	38	$-8.9768754605e-02$
14	$+1.7177623454e-01$	39	$+1.6341122499e-01$
15	$+1.2231192352e-01$	40	$-1.2656961339e-01$
16	$-2.3539333508e-02$	41	$+8.4075902816e-03$
17	$-1.4448074617e-01$	42	$+1.1751730115e-01$
18	$-8.2011766906e-02$	43	$-1.8244904376e-01$
19	$+1.0073524227e-01$	44	$+1.5796180253e-01$
20	$+1.1721629829e-01$	45	$-6.2829145363e-02$
21	$-7.8466998224e-02$	46	$-5.4315944377e-02$
22	$-1.2036615510e-01$	47	$+1.4168193497e-01$
23	$+8.9452369407e-02$	48	$-1.6594821506e-01$
24	$+1.0107656000e-01$	49	$+1.2126670027e-01$

$j = 31$**:** $\lambda_{31} = 1.334857e-13$**,** $\ell_{31}(\mathbf{1}) = -1.164209e-12$**,** $\|\hat{v}_{31} - v_{31}\|_2 \leq 2.81e-141$

k	$[v_{31}]_k$	k	$[v_{31}]_k$
0	$-3.2551689036e-14$	25	$-2.6815405982e-04$
1	$-3.9902556917e-12$	26	$+1.4079745609e-01$
2	$-1.7579466350e-10$	27	$-6.2107340380e-02$
3	$-4.3847645928e-09$	28	$-1.0748721849e-01$
4	$-7.3208344837e-08$	29	$+1.2877969136e-01$
5	$-8.9006676337e-07$	30	$+1.7251798635e-02$
6	$-8.2741193456e-06$	31	$-1.4213086812e-01$
7	$-6.0613784846e-05$	32	$+1.0289821074e-01$
8	$-3.5656926002e-04$	33	$+4.8321844372e-02$
9	$-1.7026668396e-03$	34	$-1.4976116921e-01$
10	$-6.6274090744e-03$	35	$+1.0587247171e-01$
11	$-2.0976241578e-02$	36	$+3.5055294448e-02$
12	$-5.3412826991e-02$	37	$-1.4610449742e-01$
13	$-1.0686260848e-01$	38	$+1.3885910576e-01$
14	$-1.5976294738e-01$	39	$-2.5894840807e-02$
15	$-1.5705825171e-01$	40	$-1.0479712692e-01$
16	$-5.2905162473e-02$	41	$+1.6292191318e-01$
17	$+9.9056155097e-02$	42	$-1.1506894368e-01$
18	$+1.3767568660e-01$	43	$-4.2646883278e-03$
19	$-9.3799628084e-03$	44	$+1.2406365993e-01$
20	$-1.4181232547e-01$	45	$-1.8166040430e-01$
21	$-3.2737367752e-02$	46	$+1.5334387962e-01$
22	$+1.3689481614e-01$	47	$-5.8901876835e-02$
23	$+3.3597207357e-02$	48	$-5.5078324280e-02$
24	$-1.4080436104e-01$	49	$+1.3976616736e-01$

$j = 32$**:** $\lambda_{32} = -5.083983e-14$**,** $\ell_{32}(\mathbf{1}) = +4.513023e-13$**,** $\|\hat{v}_{32} - v_{32}\|_2 \leq 1.32e-140$

k	$[v_{32}]_k$	k	$[v_{32}]_k$
0	$+1.2190371234e-14$	25	$+1.1468767512e-01$
1	$+1.5461977209e-12$	26	$-9.7410119834e-02$
2	$+7.0587793293e-11$	27	$-7.9609475009e-02$
3	$+1.8274979338e-09$	28	$+1.3353238022e-01$
4	$+3.1731580711e-08$	29	$+7.0802258777e-03$
5	$+4.0210499281e-07$	30	$-1.3789546239e-01$
6	$+3.9062067681e-06$	31	$+8.9766205103e-02$
7	$+2.9996201518e-05$	32	$+6.8336657948e-02$
8	$+1.8566522560e-04$	33	$-1.4733874950e-01$
9	$+9.3720451230e-04$	34	$+6.5244870876e-02$
10	$+3.8794344098e-03$	35	$+8.4185461974e-02$
11	$+1.3163326358e-02$	36	$-1.5123737160e-01$
12	$+3.6349063897e-02$	37	$+7.8476664021e-02$
13	$+8.0305009901e-02$	38	$+6.3016087016e-02$
14	$+1.3706333176e-01$	39	$-1.5296265512e-01$
15	$+1.6708330197e-01$	40	$+1.2439363748e-01$
16	$+1.1302079563e-01$	41	$-5.1684763526e-03$
17	$-2.8388389819e-02$	42	$-1.1661980231e-01$
18	$-1.4070150506e-01$	43	$+1.6079085391e-01$
19	$-7.9988491598e-02$	44	$-1.0409865698e-01$
20	$+9.4233819814e-02$	45	$-1.5291027077e-02$
21	$+1.1804189565e-01$	46	$+1.2924357379e-01$
22	$-6.4837517636e-02$	47	$-1.8051426773e-01$
23	$-1.2472402947e-01$	48	$+1.4925690835e-01$
24	$+6.7999969452e-02$	49	$-5.5821877285e-02$

$j = 33$**:** $\lambda_{33} = 1.936554e-14$**,** $\ell_{33}(\mathbf{1}) = -1.748680e-13$**,** $\|\hat{v}_{33} - v_{33}\|_2 \leq 6.19e-140$

k	$[v_{33}]_k$	k	$[v_{33}]_k$
0	$-4.5682661250e-15$	25	$-1.3070298261e-01$
1	$-5.9888311557e-13$	26	$-2.7300842068e-02$
2	$-2.8297346895e-11$	27	$+1.3814284443e-01$
3	$-7.5943113848e-10$	28	$-2.4352943452e-02$
4	$-1.3693379068e-08$	29	$-1.2674730402e-01$
5	$-1.8056546581e-07$	30	$+9.4071284767e-02$
6	$-1.8296218539e-06$	31	$+6.7459170590e-02$
7	$-1.4696162335e-05$	32	$-1.4153333607e-01$
8	$-9.5469604442e-05$	33	$+4.2112309270e-02$
9	$-5.0788336260e-04$	34	$+1.0614830536e-01$
10	$-2.2272251606e-03$	35	$-1.3788147235e-01$
11	$-8.0613561217e-03$	36	$+2.6365860383e-02$
12	$-2.3972247354e-02$	37	$+1.1098472765e-01$
13	$-5.7850304490e-02$	38	$-1.4478723394e-01$
14	$-1.1047635963e-01$	39	$+5.0691297731e-02$
15	$-1.5846280255e-01$	40	$+8.6016655521e-02$
16	$-1.4935575953e-01$	41	$-1.5522026481e-01$
17	$-4.4759732186e-02$	42	$+1.0906716958e-01$
18	$+9.9327135207e-02$	43	$+1.3312093394e-02$
19	$+1.3305272324e-01$	44	$-1.2574175373e-01$
20	$-7.1205836530e-03$	45	$+1.5752926423e-01$
21	$-1.3651620782e-01$	46	$-9.3819263091e-02$
22	$-4.0824698690e-02$	47	$-2.4824369439e-02$
23	$+1.2805709556e-01$	48	$+1.3329289619e-01$
24	$+4.9552210686e-02$	49	$-1.7914320566e-01$

$j = 34$**:** $\lambda_{34} = -7.377470e-15$**,** $\ell_{34}(\mathbf{1}) = +6.772837e-14$**,** $\|\hat{v}_{34} - v_{34}\|_2 \leq 2.91e-139$

k	$[v_{34}]_k$	k	$[v_{34}]_k$
0	$+1.7130036569e-15$	25	$+4.8851903818e-02$
1	$+2.3186916316e-13$	26	$+1.2227745760e-01$
2	$+1.1326584919e-11$	27	$-7.2303365394e-02$
3	$+3.1472023984e-10$	28	$-1.0100464484e-01$
4	$+5.8850036774e-09$	29	$+1.1137789277e-01$
5	$+8.0629104028e-08$	30	$+4.8057756087e-02$
6	$+8.5072584600e-07$	31	$-1.3809850749e-01$
7	$+7.1337245744e-06$	32	$+3.9483896248e-02$
8	$+4.8527262213e-05$	33	$+1.1017765310e-01$
9	$+2.7133120989e-04$	34	$-1.2470640455e-01$
10	$+1.2563626401e-03$	35	$-5.7119457905e-03$
11	$+4.8300136633e-03$	36	$+1.2983834991e-01$
12	$+1.5378220895e-02$	37	$-1.1833357861e-01$
13	$+4.0192362429e-02$	38	$-1.0343368195e-02$
14	$+8.4653159777e-02$	39	$+1.2918197396e-01$
15	$+1.3852568068e-01$	40	$-1.3276285582e-01$
16	$+1.6230918987e-01$	41	$+2.3973551623e-02$
17	$+1.0407953796e-01$	42	$+1.0434016873e-01$
18	$-3.3180194072e-02$	43	$-1.5397051868e-01$
19	$-1.3738167204e-01$	44	$+9.3611274016e-02$
20	$-7.7536619090e-02$	45	$+2.9576246814e-02$
21	$+8.9008639878e-02$	46	$-1.3262367092e-01$
22	$+1.1786997565e-01$	47	$+1.5353207078e-01$
23	$-5.3307134444e-02$	48	$-8.4318966735e-02$
24	$-1.2643115352e-01$	49	$-3.3016114441e-02$

$j = 35$**:** $\lambda_{35} = 2.810825e-15$**,** $\ell_{35}(\mathbf{1}) = -2.622160e-14$**,** $\|\hat{v}_{35} - v_{35}\|_2 \leq 1.37e-138$

k	$[v_{35}]_k$	k	$[v_{35}]_k$
0	$-6.4271834664e-16$	25	$+6.1857866203e-02$
1	$-8.9738328287e-14$	26	$-1.1919671935e-01$
2	$-4.5272053656e-12$	27	$-4.8893203425e-02$
3	$-1.3008929878e-10$	28	$+1.2951584007e-01$
4	$-2.5195132248e-09$	29	$+8.7874110836e-03$
5	$-3.5815883731e-08$	30	$-1.3296492560e-01$
6	$-3.9288275021e-07$	31	$+5.5601484445e-02$
7	$-3.4331917842e-06$	32	$+1.0159238163e-01$
8	$-2.4404978823e-05$	33	$-1.2036066427e-01$
9	$-1.4306979482e-04$	34	$-1.6407079599e-02$
10	$-6.9741931217e-04$	35	$+1.3300897311e-01$
11	$-2.8372954761e-03$	36	$-9.4704442384e-02$
12	$-9.6249056762e-03$	37	$-4.8319773707e-02$
13	$-2.7056918783e-02$	38	$+1.4061994016e-01$
14	$-6.2174097931e-02$	39	$-9.2730031538e-02$
15	$-1.1373223329e-01$	40	$-4.2901999947e-02$
16	$-1.5687190403e-01$	41	$+1.3981347906e-01$
17	$-1.4181038059e-01$	42	$-1.1712549250e-01$
18	$-3.6916985701e-02$	43	$-7.9149923439e-04$
19	$+9.9835329097e-02$	44	$+1.1846115144e-01$
20	$+1.2862863051e-01$	45	$-1.5013740494e-01$
21	$-5.8572804820e-03$	46	$+7.8526126672e-02$
22	$-1.3172728187e-01$	47	$+4.3742103579e-02$
23	$-4.6916020212e-02$	48	$-1.3767160339e-01$
24	$+1.1953227248e-01$	49	$+1.4910044462e-01$

$j = 36$**:** $\lambda_{36} = -1.071039e-15$**,** $\ell_{36}(\mathbf{1}) = +1.014812e-14$**,** $\|\hat{v}_{36} - v_{36}\|_2 \leq 6.44e-138$

k	$[v_{36}]_k$	k	$[v_{36}]_k$
0	$+2.4128104872e-16$	25	$-1.2634011528e-01$
1	$+3.4718129081e-14$	26	$+3.2119837843e-02$
2	$+1.8070762352e-12$	27	$+1.2545516926e-01$
3	$+5.3642366992e-11$	28	$-4.8652278268e-02$
4	$+1.0747954517e-09$	29	$-1.1411387124e-01$
5	$+1.5831960577e-08$	30	$+8.5671057785e-02$
6	$+1.8029399942e-07$	31	$+7.8807089214e-02$
7	$+1.6391244535e-06$	32	$-1.2448189618e-01$
8	$+1.2153036565e-05$	33	$-7.9067922250e-03$
9	$+7.4533717198e-05$	34	$+1.2951440797e-01$
10	$+3.8149057123e-04$	35	$-8.3920420310e-02$
11	$+1.6370338395e-03$	36	$-6.4613654270e-02$
12	$+5.8920358818e-03$	37	$+1.3774076823e-01$
13	$+1.7713879526e-02$	38	$-5.8059823587e-02$
14	$+4.4034713263e-02$	39	$-8.2881847408e-02$
15	$+8.8751377256e-02$	40	$+1.4068666768e-01$
16	$+1.3961393726e-01$	41	$-6.4286150400e-02$
17	$+1.5744365866e-01$	42	$-7.0366215039e-02$
18	$+9.5438458637e-02$	43	$+1.4414036016e-01$
19	$-3.7881174041e-02$	44	$-9.9423605015e-02$
20	$-1.3440817913e-01$	45	$-2.3132758737e-02$
21	$-7.4764400376e-02$	46	$+1.2893184312e-01$
22	$+8.4849390899e-02$	47	$-1.4447723807e-01$
23	$+1.1702823465e-01$	48	$+6.4138142792e-02$
24	$-4.3632146860e-02$	49	$+5.5975488227e-02$

$j = 37$**:** $\lambda_{37} = 4.081489e-16$**,** $\ell_{37}(\mathbf{1}) = -3.926074e-15$**,** $\|\hat{v}_{37} - v_{37}\|_2 \leq 3.03e-137$

k	$[v_{37}]_k$	k	$[v_{37}]_k$
0	$-9.0625781893e-17$	25	$+1.1156667145e-01$
1	$-1.3427233587e-14$	26	$+7.1227933709e-02$
2	$-7.2039669012e-13$	27	$-1.0728792952e-01$
3	$-2.2069190830e-11$	28	$-6.5676517275e-02$
4	$-4.5695138276e-10$	29	$+1.1732922822e-01$
5	$-6.9663167620e-09$	30	$+3.6375666470e-02$
6	$-8.2247865564e-08$	31	$-1.2959879342e-01$
7	$-7.7677177242e-07$	32	$+1.8395916106e-02$
8	$-5.9966565317e-06$	33	$+1.2033234206e-01$
9	$-3.8397807889e-05$	34	$-8.7580013585e-02$
10	$-2.0587017355e-04$	35	$-6.3839981814e-02$
11	$-9.2913550260e-04$	36	$+1.3291146099e-01$
12	$-3.5352925458e-03$	37	$-3.8861518942e-02$
13	$-1.1312333160e-02$	38	$-1.0056695600e-01$
14	$-3.0214569309e-02$	39	$+1.2807301715e-01$
15	$-6.6373657346e-02$	40	$-1.9854834349e-02$
16	$-1.1664270574e-01$	41	$-1.0841751495e-01$
17	$-1.5500233648e-01$	42	$+1.3251664643e-01$
18	$-1.3438961590e-01$	43	$-3.5375303043e-02$
19	$-2.9354365982e-02$	44	$-9.2496995799e-02$
20	$+1.0049482557e-01$	45	$+1.4343897622e-01$
21	$+1.2436604033e-01$	46	$-8.0827298570e-02$
22	$-5.3867097930e-03$	47	$-4.2865694693e-02$
23	$-1.2744019296e-01$	48	$+1.3631291547e-01$
24	$-5.1433916338e-02$	49	$-1.3759408215e-01$

$j = 38$**:** $\lambda_{38} = -1.555505e-16$**,** $\ell_{38}(\mathbf{1}) = +1.518400e-15$**,** $\|\hat{v}_{38} - v_{38}\|_2 \leq 1.43e-136$

k	$[v_{38}]_k$	k	$[v_{38}]_k$
0	$+3.4055987834e-17$	25	$-3.5571778740e-02$
1	$+5.1913046270e-15$	26	$-1.2506302220e-01$
2	$+2.8684475539e-13$	27	$+1.7710264901e-02$
3	$+9.0601252115e-12$	28	$+1.2549950206e-01$
4	$+1.9365838794e-10$	29	$-2.7241196846e-02$
5	$+3.0521304903e-09$	30	$-1.2068560435e-01$
6	$+3.7312628357e-08$	31	$+5.9339907804e-02$
7	$+3.6555648781e-07$	32	$+9.9741204210e-02$
8	$+2.9337544704e-06$	33	$-1.0248794550e-01$
9	$+1.9577498203e-05$	34	$-4.7563988730e-02$
10	$+1.0971610118e-04$	35	$+1.3005872357e-01$
11	$+5.1945917052e-04$	36	$-3.6710592099e-02$
12	$+2.0828286551e-03$	37	$-1.0390018225e-01$
13	$+7.0643540577e-03$	38	$+1.1694132567e-01$
14	$+2.0159050116e-02$	39	$+6.8612641965e-03$
15	$+4.7860739203e-02$	40	$-1.2297939347e-01$
16	$+9.2599783044e-02$	41	$+1.0821856135e-01$
17	$+1.4034529520e-01$	42	$+1.6371502818e-02$
18	$+1.5248052519e-01$	43	$-1.2514305622e-01$
19	$+8.7060332743e-02$	44	$+1.1847366201e-01$
20	$-4.2465557250e-02$	45	$-7.6282779317e-03$
21	$-1.3169122088e-01$	46	$-1.0949931610e-01$
22	$-7.1747772458e-02$	47	$+1.3888681839e-01$
23	$+8.1580526799e-02$	48	$-6.2185574266e-02$
24	$+1.1574130539e-01$	49	$-5.9995246138e-02$

$j = 39$**:** $\lambda_{39} = 5.928725e-17$**,** $\ell_{39}(\mathbf{1}) = -5.870515e-16$**,** $\|\hat{v}_{39} - v_{39}\|_2 \leq 6.72e-136$

k	$[v_{39}]_k$	k	$[v_{39}]_k$
0	$-1.2803730896e-17$	25	$-5.4705982137e-02$
1	$-2.0064736299e-15$	26	$+1.0428071693e-01$
2	$-1.1408552621e-13$	27	$+7.8263106326e-02$
3	$-3.7119631555e-12$	28	$-9.5603632037e-02$
4	$-8.1828450951e-11$	29	$-7.8386898078e-02$
5	$-1.3318192165e-09$	30	$+1.0331808153e-01$
6	$-1.6839242970e-08$	31	$+5.8371667849e-02$
7	$-1.7091750112e-07$	32	$-1.1971396409e-01$
8	$-1.4238793287e-06$	33	$-1.4705086670e-02$
9	$-9.8858943099e-06$	34	$+1.2617912603e-01$
10	$-5.7797716042e-05$	35	$-5.0409664003e-02$
11	$-2.8640756212e-04$	36	$-9.6634568453e-02$
12	$-1.2067547524e-03$	37	$+1.1329940199e-01$
13	$-4.3229658195e-03$	38	$+1.5827316135e-02$
14	$-1.3117538591e-02$	39	$-1.2549658592e-01$
15	$-3.3430435912e-02$	40	$+8.8072185884e-02$
16	$-7.0439319487e-02$	41	$+4.7876162550e-02$
17	$-1.1921819808e-01$	42	$-1.3265463010e-01$
18	$-1.5286526226e-01$	43	$+8.2105119932e-02$
19	$-1.2706979463e-01$	44	$+4.8429665570e-02$
20	$-2.2054951850e-02$	45	$-1.3396065920e-01$
21	$+1.0123864035e-01$	46	$+1.0061122118e-01$
22	$+1.2023084367e-01$	47	$+1.7924530721e-02$
23	$-5.5523841307e-03$	48	$-1.2183172515e-01$
24	$-1.2362646341e-01$	49	$+1.3150976816e-01$

$j = 40$**:** $\lambda_{40} = -2.259881e-17$**,** $\ell_{40}(\mathbf{1}) = +2.269005e-16$**,** $\|\hat{v}_{40} - v_{40}\|_2 \leq 3.17e-135$

k	$[v_{40}]_k$	k	$[v_{40}]_k$
0	$+4.8158179342e-18$	25	$+1.1416288121e-01$
1	$+7.7528976862e-16$	26	$-2.8904331793e-02$
2	$+4.5326146309e-14$	27	$-1.2303565459e-01$
3	$+1.5178988339e-12$	28	$+5.4309868025e-03$
4	$+3.4478382631e-11$	29	$+1.2339223578e-01$
5	$+5.7893664767e-10$	30	$-8.3756607715e-03$
6	$+7.5623920114e-09$	31	$-1.2232034702e-01$
7	$+7.9426070713e-08$	32	$+3.4211008406e-02$
8	$+6.8592736401e-07$	33	$+1.1213394301e-01$
9	$+4.9472239700e-06$	34	$-7.6395078570e-02$
10	$+3.0120825200e-05$	35	$-7.7590077310e-02$
11	$+1.5589326944e-04$	36	$+1.1694065737e-01$
12	$+6.8849888492e-04$	37	$+8.5588486092e-03$
13	$+2.5968895700e-03$	38	$-1.2250833981e-01$
14	$+8.3451919930e-03$	39	$+7.8944722462e-02$
15	$+2.2702694086e-02$	40	$+6.1570367948e-02$
16	$+5.1656232029e-02$	41	$-1.2980394545e-01$
17	$+9.6198138048e-02$	42	$+5.2430238553e-02$
18	$+1.4073463568e-01$	43	$+8.1142089296e-02$
19	$+1.4741676145e-01$	44	$-1.3151892658e-01$
20	$+7.8917330186e-02$	45	$+5.3002895281e-02$
21	$-4.6913057891e-02$	46	$+7.5186059860e-02$
22	$-1.2915888361e-01$	47	$-1.3609575367e-01$
23	$-6.8540887732e-02$	48	$+8.0603144415e-02$
24	$+7.9055876811e-02$	49	$+4.0709434272e-02$

$j = 41$**:** $\lambda_{41} = 8.614744e-18$**,** $\ell_{41}(\mathbf{1}) = -8.767390e-17$**,** $\|\hat{v}_{41} - v_{41}\|_2 \leq 1.49e-134$

k	$[v_{41}]_k$	k	$[v_{41}]_k$
0	$-1.8121077708e-18$	25	$-1.2024628014e-01$
1	$-2.9948459538e-16$	26	$-5.6986055656e-02$
2	$-1.7989777380e-14$	27	$+9.7717778387e-02$
3	$-6.1957788797e-13$	28	$+8.3455580036e-02$
4	$-1.4488747047e-11$	29	$-8.4516930151e-02$
5	$-2.5075703089e-10$	30	$-8.7744702462e-02$
6	$-3.3805578679e-09$	31	$+8.8676264693e-02$
7	$-3.6697962435e-08$	32	$+7.5217614033e-02$
8	$-3.2812434047e-07$	33	$-1.0579177374e-01$
9	$-2.4549642085e-06$	34	$-4.2386403130e-02$
10	$-1.5540116643e-05$	35	$+1.2217515511e-01$
11	$-8.3845216439e-05$	36	$-1.3868062831e-02$
12	$-3.8727161271e-04$	37	$-1.1480090200e-01$
13	$-1.5337474767e-03$	38	$+8.2130681757e-02$
14	$-5.2014537978e-03$	39	$+6.0848334221e-02$
15	$-1.5034106638e-02$	40	$-1.2564586897e-01$
16	$-3.6690528965e-02$	41	$+3.6079847711e-02$
17	$-7.4362146081e-02$	42	$+9.5876130279e-02$
18	$-1.2146759467e-01$	43	$-1.2000809351e-01$
19	$-1.5047124589e-01$	44	$+1.5010095318e-02$
20	$-1.1983397666e-01$	45	$+1.0542459452e-01$
21	$-1.5006209557e-02$	46	$-1.2193595304e-01$
22	$+1.0201386270e-01$	47	$+2.3417266541e-02$
23	$+1.1619282771e-01$	48	$+9.6267960414e-02$
24	$-6.2323517485e-03$	49	$-1.3285976766e-01$

$j = 42$**:** $\lambda_{42} = -3.284201e-18$**,** $\ell_{42}(\mathbf{1}) = +3.386778e-17$**,** $\|\hat{v}_{42} - v_{42}\|_2 \leq 7.03e-134$

k	$[v_{42}]_k$	k	$[v_{42}]_k$
0	$+6.8213316410e-19$	25	$+7.7153596724e-02$
1	$+1.1565662134e-16$	26	$+1.1239760588e-01$
2	$+7.1331906001e-15$	27	$-2.3433251882e-02$
3	$+2.5246644037e-13$	28	$-1.2056555095e-01$
4	$+6.0732012723e-12$	29	$-4.9485371598e-03$
5	$+1.0824220185e-10$	30	$+1.1987038018e-01$
6	$+1.5046046698e-09$	31	$+7.9241056653e-03$
7	$+1.6864254281e-08$	32	$-1.2037194219e-01$
8	$+1.5593217901e-07$	33	$+1.1310626144e-02$
9	$+1.2086313156e-06$	34	$+1.1756782132e-01$
10	$+7.9424832946e-06$	35	$-4.9286824130e-02$
11	$+4.4595633210e-05$	36	$-9.7923549796e-02$
12	$+2.1498385135e-04$	37	$+9.5150954294e-02$
13	$+8.9179011220e-04$	38	$+4.6949830010e-02$
14	$+3.1819052345e-03$	39	$-1.2324865755e-01$
15	$+9.7322157544e-03$	40	$+3.4307252973e-02$
16	$+2.5333962450e-02$	41	$+9.8642711878e-02$
17	$+5.5407953031e-02$	42	$-1.1036462612e-01$
18	$+9.9545961502e-02$	43	$-7.7855754760e-03$
19	$+1.4079513827e-01$	44	$+1.1714816340e-01$
20	$+1.4225183610e-01$	45	$-1.0001944433e-01$
21	$+7.0988802181e-02$	46	$-2.0555494266e-02$
22	$-5.1207344196e-02$	47	$+1.2073984798e-01$
23	$-1.2675318702e-01$	48	$-1.0627392306e-01$
24	$-6.5183047602e-02$	49	$-4.8724871601e-03$

$j = 43$: $\lambda_{43} = 1.252120e-18$, $\quad \ell_{43}(\mathbf{1}) = -1.307948e-17$, $\quad \|\hat{v}_{43} - v_{43}\|_2 \leq 3.32e-133$

k	$[v_{43}]_k$	k	$[v_{43}]_k$
0	$-2.5687208536e-19$	25	$-7.3303783665e-03$
1	$-4.4653697179e-17$	26	$-1.1725529860e-01$
2	$-2.8258139309e-15$	27	$-5.8471106571e-02$
3	$-1.0270787030e-13$	28	$+9.1875226577e-02$
4	$-2.5395931000e-12$	29	$+8.7201878519e-02$
5	$-4.6573604414e-11$	30	$-7.4232050016e-02$
6	$-6.6690552994e-10$	31	$-9.4400039244e-02$
7	$-7.7102842113e-09$	32	$+7.4191444137e-02$
8	$-7.3643696655e-08$	33	$+8.7571218432e-02$
9	$-5.9062756410e-07$	34	$-8.9714563310e-02$
10	$-4.0237293257e-06$	35	$-6.4328756730e-02$
11	$-2.3474033631e-05$	36	$+1.1125507003e-01$
12	$-1.1788891638e-04$	37	$+1.8951252861e-02$
13	$-5.1107936888e-04$	38	$-1.2025107333e-01$
14	$-1.9133192783e-03$	39	$+4.6140945706e-02$
15	$-6.1713299795e-03$	40	$+9.2402012144e-02$
16	$-1.7055351315e-02$	41	$-1.0727480272e-01$
17	$-3.9981516638e-02$	42	$-1.5405798512e-02$
18	$-7.8133817347e-02$	43	$+1.1913292545e-01$
19	$-1.2339864144e-01$	44	$-8.2486919636e-02$
20	$-1.4783048391e-01$	45	$-4.7334541795e-02$
21	$-1.1267042868e-01$	46	$+1.2587379236e-01$
22	$-8.1990408253e-03$	47	$-7.3661960024e-02$
23	$+1.0277823430e-01$	48	$-5.1915464984e-02$
24	$+1.1222576235e-01$	49	$+1.2787631400e-01$

$j = 44$: $\lambda_{44} = -4.774076e-19$, $\quad \ell_{44}(\mathbf{1}) = +5.049939e-18$, $\quad \|\hat{v}_{44} - v_{44}\|_2 \leq 1.56e-132$

k	$[v_{44}]_k$	k	$[v_{44}]_k$
0	$+9.6765225049e-20$	25	$-6.1703344125e-02$
1	$+1.7236148126e-17$	26	$+7.5771801027e-02$
2	$+1.1184694903e-15$	27	$+1.1051604318e-01$
3	$+4.1718720253e-14$	28	$-1.8987973470e-02$
4	$+1.0595528951e-12$	29	$-1.1786762446e-01$
5	$+1.9978139206e-11$	30	$-1.3663629541e-02$
6	$+2.9444820474e-10$	31	$+1.1547719083e-01$
7	$+3.5081046822e-09$	32	$+2.1797681456e-02$
8	$+3.4577159451e-08$	33	$-1.1593244631e-01$
9	$+2.8661073314e-07$	34	$-8.8796220830e-03$
10	$+2.0216378896e-06$	35	$+1.1761648314e-01$
11	$+1.2236323699e-05$	36	$-2.3206798360e-02$
12	$+6.3910701110e-05$	37	$-1.0953099413e-01$
13	$+2.8898825552e-04$	38	$+6.8881229825e-02$
14	$+1.1324084682e-03$	39	$+7.6192772508e-02$
15	$+3.8401243373e-03$	40	$-1.1052704182e-01$
16	$+1.1222591137e-02$	41	$-9.1013626576e-03$
17	$+2.8042174968e-02$	42	$+1.1655081860e-01$
18	$+5.9103487027e-02$	43	$-7.4654222673e-02$
19	$+1.0264263590e-01$	44	$-5.9001693870e-02$
20	$+1.4053873591e-01$	45	$+1.2304498805e-01$
21	$+1.3698722329e-01$	46	$-4.7791689677e-02$
22	$+6.3259655630e-02$	47	$-7.9382983459e-02$
23	$-5.5335005413e-02$	48	$+1.2375963113e-01$
24	$-1.2442713148e-01$	49	$-4.4239723203e-02$

$j = 45$**:** $\lambda_{45} = 1.820364e-19$**,** $\ell_{45}(\mathbf{1}) = -1.949302e-18$**,** $\|\hat{v}_{45} - v_{45}\|_2 \leq 7.38e-132$

k	$[v_{45}]_k$	k	$[v_{45}]_k$
0	$-3.6464391843e-20$	25	$+1.0830721750e-01$
1	$-6.6515600370e-18$	26	$-8.7695665814e-03$
2	$-4.4232691798e-16$	27	$-1.1460851863e-01$
3	$-1.6920678118e-14$	28	$-5.9314351637e-02$
4	$-4.4110474581e-13$	29	$+8.6724001163e-02$
5	$-8.5449380907e-12$	30	$+8.9818019020e-02$
6	$-1.2952201681e-10$	31	$-6.4843275114e-02$
7	$-1.5888568426e-09$	32	$-9.8912895933e-02$
8	$-1.6144913664e-08$	33	$+6.0356833291e-02$
9	$-1.3816609766e-07$	34	$+9.6150171387e-02$
10	$-1.0078383088e-06$	35	$-7.2840416950e-02$
11	$-6.3203490854e-06$	36	$-8.0802421022e-02$
12	$-3.4278960456e-05$	37	$+9.5933356903e-02$
13	$-1.6137550135e-04$	38	$+4.6449050159e-02$
14	$-6.6045221973e-04$	39	$-1.1569311707e-01$
15	$-2.3484016825e-03$	40	$+1.0318283601e-02$
16	$-7.2326419947e-03$	41	$+1.0997245139e-01$
17	$-1.9174350563e-02$	42	$-7.7461355722e-02$
18	$-4.3290636323e-02$	43	$-5.8337696270e-02$
19	$-8.1746562389e-02$	44	$+1.1942750203e-01$
20	$-1.2501825496e-01$	45	$-3.3680141764e-02$
21	$-1.4495296918e-01$	46	$-9.1826203942e-02$
22	$-1.0557149576e-01$	47	$+1.1315918646e-01$
23	$-1.6270479417e-03$	48	$-1.1177177539e-02$
24	$+1.0349769628e-01$	49	$-1.0250139563e-01$

$j = 46$**:** $\lambda_{46} = -6.941474e-20$**,** $\ell_{46}(\mathbf{1}) = +7.522695e-19$**,** $\|\hat{v}_{46} - v_{46}\|_2 \leq 3.48e-131$

k	$[v_{46}]_k$	k	$[v_{46}]_k$
0	$+1.3745452013e-20$	25	$-1.2214247149e-01$
1	$+2.5663249160e-18$	26	$-5.8123807757e-02$
2	$+1.7479049676e-16$	27	$+7.4824912988e-02$
3	$+6.8532116748e-15$	28	$+1.0856499971e-01$
4	$+1.8325947748e-13$	29	$-1.5422327161e-02$
5	$+3.6447129041e-12$	30	$-1.1509012861e-01$
6	$+5.6773957589e-11$	31	$-2.0936959961e-02$
7	$+7.1648901251e-10$	32	$+1.1060672870e-01$
8	$+7.4989835133e-09$	33	$+3.3464624156e-02$
9	$+6.6190838468e-08$	34	$-1.0985340360e-01$
10	$+4.9875064610e-07$	35	$-2.6230916871e-02$
11	$+3.2366305514e-06$	36	$+1.1369077725e-01$
12	$+1.8202191561e-05$	37	$+6.1312602032e-04$
13	$+8.9067183950e-05$	38	$-1.1384127669e-01$
14	$+3.7997132017e-04$	39	$+4.1311124674e-02$
15	$+1.4133048222e-03$	40	$+9.5869907991e-02$
16	$+4.5733687766e-03$	41	$-8.8970089990e-02$
17	$+1.2813030823e-02$	42	$-4.6290081716e-02$
18	$+3.0816803386e-02$	43	$+1.1737985860e-01$
19	$+6.2731135266e-02$	44	$-3.2186508228e-02$
20	$+1.0548749864e-01$	45	$-9.4172376779e-02$
21	$+1.3997645756e-01$	46	$+1.0471536340e-01$
22	$+1.3162602902e-01$	47	$+8.5238957340e-03$
23	$+5.5719138175e-02$	48	$-1.1206289962e-01$
24	$-5.9284844754e-02$	49	$+9.3131256057e-02$

$j = 47$**:** $\lambda_{47} = 2.647086e-20$**,** $\ell_{47}(\mathbf{1}) = -2.902510e-19$**,** $\|\hat{v}_{47} - v_{47}\|_2 \leq 1.64e-130$

k	$[v_{47}]_k$	k	$[v_{47}]_k$
0	$-5.1830220205e-21$	25	$+1.0414460268e-01$
1	$-9.8993946689e-19$	26	$+1.0441826271e-01$
2	$-6.9018090391e-17$	27	$-1.0487694706e-02$
3	$-2.7719673785e-15$	28	$-1.1226241839e-01$
4	$-7.5986887043e-14$	29	$-5.9635347769e-02$
5	$-1.5505137057e-12$	30	$+8.2220879855e-02$
6	$-2.4802784531e-11$	31	$+9.1554027465e-02$
7	$-3.2176701874e-10$	32	$-5.6375011890e-02$
8	$-3.4658270624e-09$	33	$-1.0175168246e-01$
9	$-3.1522879339e-08$	34	$+4.7458547183e-02$
10	$-2.4510777951e-07$	35	$+1.0164632685e-01$
11	$-1.6440735660e-06$	36	$-5.6102762416e-02$
12	$-9.5746504520e-06$	37	$-9.2388369359e-02$
13	$-4.8622971565e-05$	38	$+7.8209611828e-02$
14	$-2.1583927654e-04$	39	$+6.8071712986e-02$
15	$-8.3799493378e-04$	40	$-1.0393196879e-01$
16	$-2.8416451525e-03$	41	$-2.2078504266e-02$
17	$-8.3849452961e-03$	42	$+1.1502800030e-01$
18	$-2.1383974222e-02$	43	$-4.2550912506e-02$
19	$-4.6605627229e-02$	44	$-8.8735650778e-02$
20	$-8.5193114486e-02$	45	$+1.0204809177e-01$
21	$-1.2633276449e-01$	46	$+1.5107531162e-02$
22	$-1.4184860887e-01$	47	$-1.1366848947e-01$
23	$-9.8532750955e-02$	48	$+7.7718289123e-02$
24	$+4.7140404403e-03$	49	$+4.6717100523e-02$

$j = 48$**:** $\lambda_{48} = -1.009500e-20$**,** $\ell_{48}(\mathbf{1}) = +1.119653e-19$**,** $\|\hat{v}_{48} - v_{48}\|_2 \leq 7.76e-130$

k	$[v_{48}]_k$	k	$[v_{48}]_k$
0	$+1.9549473828e-21$	25	$-6.3047389138e-02$
1	$+3.8178450845e-19$	26	$-1.1986801850e-01$
2	$+2.7232854922e-17$	27	$-5.4461567168e-02$
3	$+1.1197621309e-15$	28	$+7.4240673941e-02$
4	$+3.1448312568e-14$	29	$+1.0657469089e-01$
5	$+6.5795941258e-13$	30	$-1.2611937046e-02$
6	$+1.0801071412e-11$	31	$-1.1233348232e-01$
7	$+1.4393566375e-10$	32	$-2.6970434842e-02$
8	$+1.5942549582e-09$	33	$+1.0554019483e-01$
9	$+1.4928574072e-08$	34	$+4.3175615250e-02$
10	$+1.1966665388e-07$	35	$-1.0278152008e-01$
11	$+8.2874522910e-07$	36	$-4.0829647022e-02$
12	$+4.9918647475e-06$	37	$+1.0697863400e-01$
13	$+2.6272335774e-05$	38	$+2.1533866628e-02$
14	$+1.2115395913e-04$	39	$-1.1239592731e-01$
15	$+4.9004564081e-04$	40	$+1.4656163488e-02$
16	$+1.7372880195e-03$	41	$+1.0669003746e-01$
17	$+5.3830460372e-03$	42	$-6.2665851601e-02$
18	$+1.4499833943e-02$	43	$-7.4737356576e-02$
19	$+3.3647458961e-02$	44	$+1.0498442188e-01$
20	$+6.6279836883e-02$	45	$+9.5664138661e-03$
21	$+1.0807992113e-01$	46	$-1.1140292274e-01$
22	$+1.3911868905e-01$	47	$+7.0893792005e-02$
23	$+1.2617270160e-01$	48	$+5.6813250795e-02$
24	$+4.8359915560e-02$	49	$-1.1721657727e-01$

$j = 49$**:** $\lambda_{49} = 3.850039e-21, \quad \ell_{49}(\mathbf{1}) = -4.318241e-20, \quad \|\hat{v}_{49} - v_{49}\|_2 \leq 3.67e-129$

k	$[v_{49}]_k$	k	$[v_{49}]_k$
0	$-7.3758060019e-22$	25	$+1.0826823066e-02$
1	$-1.4721233018e-19$	26	$+1.0469640064e-01$
2	$-1.0737965942e-17$	27	$+1.0054312788e-01$
3	$-4.5178469510e-16$	28	$-1.2433779041e-02$
4	$-1.2992086239e-14$	29	$-1.1017605640e-01$
5	$-2.7853737158e-13$	30	$-5.9527721262e-02$
6	$-4.6893527300e-12$	31	$+7.8316082656e-02$
7	$-6.4146289312e-11$	32	$+9.2606704498e-02$
8	$-7.3005497636e-10$	33	$-4.8808728001e-02$
9	$-7.0323074730e-09$	34	$-1.0330045406e-01$
10	$-5.8060415150e-08$	35	$+3.5640825386e-02$
11	$-4.1473605420e-07$	36	$+1.0468366235e-01$
12	$-2.5808327752e-06$	37	$-4.0108724312e-02$
13	$-1.4058920144e-05$	38	$-9.9796635173e-02$
14	$-6.7250200388e-05$	39	$+5.9583692232e-02$
15	$-2.8289142450e-04$	40	$+8.3938310178e-02$
16	$-1.0462764988e-03$	41	$-8.7497834450e-02$
17	$-3.3954138064e-03$	42	$-4.9226783080e-02$
18	$-9.6273325470e-03$	43	$+1.0999198153e-01$
19	$-2.3676907698e-02$	44	$-7.4625644434e-03$
20	$-4.9914679595e-02$	45	$-1.0571458797e-01$
21	$-8.8466683314e-02$	46	$+7.3398323310e-02$
22	$-1.2734810290e-01$	47	$+5.6196323014e-02$
23	$-1.3852730907e-01$	48	$-1.1402183084e-01$
24	$-9.1552344183e-02$	49	$+3.1567831525e-02$

$j = 50$**:** $\lambda_{50} = -1.468398e-21, \quad \ell_{50}(\mathbf{1}) = +1.665125e-20, \quad \|\hat{v}_{50} - v_{50}\|_2 \leq 1.73e-128$

k	$[v_{50}]_k$	k	$[v_{50}]_k$
0	$+2.7835629669e-22$	25	$+4.1177360547e-02$
1	$+5.6753106159e-20$	26	$-6.6614545461e-02$
2	$+4.2311862301e-18$	27	$-1.1757833392e-01$
3	$+1.8206570357e-16$	28	$-5.0730345707e-02$
4	$+5.3581796698e-15$	29	$+7.3957717244e-02$
5	$+1.1764542595e-13$	30	$+1.0456374951e-01$
6	$+2.0300126303e-12$	31	$-1.0451352936e-02$
7	$+2.8485674436e-11$	32	$-1.0966386661e-01$
8	$+3.3288442366e-10$	33	$-3.1942392723e-02$
9	$+3.2959293692e-09$	34	$+1.0047453126e-01$
10	$+2.8003675784e-08$	35	$+5.1183706820e-02$
11	$+2.0612860068e-07$	36	$-9.5196677880e-02$
12	$+1.3237690384e-06$	37	$-5.2883204656e-02$
13	$+7.4548749103e-06$	38	$+9.8437726306e-02$
14	$+3.6939313233e-05$	39	$+3.9324624158e-02$
15	$+1.6134225449e-04$	40	$-1.0665078190e-01$
16	$+6.2135557393e-04$	41	$-9.6781119087e-03$
17	$+2.1070097985e-03$	42	$+1.0994674614e-01$
18	$+6.2701608076e-03$	43	$-3.4856680135e-02$
19	$+1.6278920016e-02$	44	$-9.3744041071e-02$
20	$+3.6523884116e-02$	45	$+8.3673593441e-02$
21	$+6.9739111552e-02$	46	$+4.5625655594e-02$
22	$+1.1041937685e-01$	47	$-1.1225370696e-01$
23	$+1.3797537308e-01$	48	$+3.0294155178e-02$
24	$+1.2063280321e-01$	49	$+9.0312194250e-02$

Tail Bound for the Spectral Remainder

The spectral expansion (1) has remainder

$$R_N(n) \;=\; L^n\mathbf{1} - \sum_{j=1}^{N} \lambda_j^n\, \ell_j(\mathbf{1})\, v_j.$$

We bound $\|R_N(n)\|_2$ by combining the Cauchy integral formula with a direct computation of the *tail projection* $Q_N\mathbf{1}$, where $Q_N = I - \sum_{j=1}^{N} P_j$ is the tail spectral projector.

Factored tail bound. Since Q_N commutes with L_r (Riesz projectors commute with the operator, regardless of normality), we have the factorization

$$R_N(n) = Q_N L^n\mathbf{1} = (Q_N L^n)(Q_N\mathbf{1}),$$

using only $Q_N^2 = Q_N$ (Riesz idempotency) and L-invariance of range(Q_N). The Cauchy integral on a circle $\Gamma = \{|z| = \rho\}$ in the eigenvalue gap $|\lambda_{N+1}| < \rho < |\lambda_N|$ bounds the operator factor:

$$\|Q_N L^n\| = \left\| \frac{1}{2\pi i} \oint_\Gamma z^n R_{L_r}(z)\, dz \right\| \le \rho^{n+1} \cdot M_\infty,$$

where $M_\infty = \sup_{z\in\Gamma} \|R_{L_r}(z)\|$ is obtained via the resolvent bridge. The tail projection $\|Q_N\mathbf{1}\|_2 = \|\mathbf{1} - \sum_{j=1}^{N} P_j\mathbf{1}\|_2$ is computed directly from the rigorous spectral coefficients.

The combined bound is:

$$\boxed{\|R_N(n)\|_2 \le \rho^{n+1} \cdot M_\infty \cdot \|Q_N\mathbf{1}\|_2.}$$

Tail bound at $N = 50$. Using all $N = 50$ certified eigenvalues, the separating contour Γ lies in the eigenvalue gap $|\lambda_{51}| < \rho < |\lambda_{50}|$, with $|\lambda_{50}| \approx 1.47 \times 10^{-21}$ and $|\lambda_{51}| \approx 5.60 \times 10^{-22}$. The resolvent on this circle is certified via the block Schur method from the ordered Schur form at $K = 256$ (Phase 3 of the full certification pipeline). The tail projection $Q_{50}\mathbf{1}$ is computed at $K = 512$ (1024-bit precision) using the certified $\ell_j(\mathbf{1})$ and Schur eigenvectors; the correction for the infinite-dimensional lift ($N \cdot \varepsilon_{512}/\mathrm{sep}_{\min} \approx 4.37 \times 10^{-68}$) is negligible compared to the Galerkin norm.

Table 6: Tail bound parameters for the spectral remainder $R_{50}(n)$. Resolvent certified at $K = 256$; tail projection $\|Q_{50}\mathbf{1}\|$ computed at $K = 512$.

Parameter	Value
$\lvert\lambda_{50}\rvert$	1.468398×10^{-21}
$\lvert\lambda_{51}\rvert$	5.600685×10^{-22}
Circle radius ρ	1.014211×10^{-21}
$M_\infty = \|R_{L_r}\|$ on Γ	6.0617×10^{41}
$\|Q_{50}\mathbf{1}\|_2$	6.1383×10^{-21}
Prefactor $C = M_\infty \cdot \|Q_{50}\mathbf{1}\|$	3.7208×10^{21}

Comparison with $N = 20$. For comparison, the tail bound at $N = 20$ (resolvent certified at $K = 64$, tail projection at $K = 256$) gives $\rho_{20} = 3.81\times10^{-9}$, $M_\infty = 3.61\times10^{9}$, $\|Q_{20}\mathbf{1}\| = 3.64\times10^{-1}$, and prefactor $C_{20} = 1.32 \times 10^{9}$. The $N = 50$ tail circle is 10^{12} times smaller in radius, yielding bounds that are stronger by a factor of $\sim 10^{12(n+1)}$.

Table 7: Rigorous upper bounds on $\|R_{50}(n)\|_2 \le \rho^{n+1} \cdot M_\infty \cdot \|Q_{50}\mathbf{1}\|_2$.

n	$\|R_{50}(n)\|_2 \le$
1	$< 10^{-20}$
2	$< 10^{-41}$
5	$< 10^{-104}$
10	$< 10^{-209}$
20	$< 10^{-419}$
50	$< 10^{-1049}$
100	$< 10^{-2099}$

Integrated Spectral Expansion

The tail bound provides a rigorous L^∞ certificate for the integrated spectral expansion. Define

$$F_n^{(j_0)}(x) \;=\; \int_0^x \sum_{j=j_0}^{N} \lambda_j^n \, \ell_j(\mathbf{1}) \, v_j(t) \, dt,$$

which represents the contribution of modes $j \ge j_0$ to the antiderivative of $L^n\mathbf{1}$. For $j_0 = 2$, this measures the integrated deviation of $L^n\mathbf{1}$ from its equilibrium $\ell_1(\mathbf{1})\, v_1$. Removing additional dominant eigenfunctions reveals progressively finer structure linked to the Gauss map's Markov partition.

The integration $\int_0^x$ can be computed exactly in the shifted monomial basis: $\int_0^x (t-1)^k \, dt = ((x-1)^{k+1} - (-1)^{k+1})/(k+1)$. The L^∞ error bound follows from Cauchy–Schwarz and the Gram matrix of the shifted monomial basis $\{(x-1)^k\}$ on $[0,1]$ (whose spectral norm equals that of the Hilbert matrix, $\le \pi$):

$$\left| F_n^{(j_0)}(x) - \int_0^x \Big(L^n\mathbf{1}(t) - \sum_{j=1}^{j_0-1} \lambda_j^n \, P_j\mathbf{1}(t) \Big) dt \right| \;\le\; \sqrt{\pi}\, \|R_N(n)\|_2 \;\le\; \sqrt{\pi}\, \rho^{n+1} \cdot C.$$

Table 8: Signal amplitude $\max_x |F_n^{(j_0)}(x)|$ by mode cutoff j_0 and iterate n, with rigorous L^∞ error bounds.

j_0	Dominant λ_{j_0}	$n=1$	$n=2$	$n=5$
2	-0.3037	3.0×10^{-2}	8.7×10^{-3}	2.5×10^{-4}
3	0.1009	1.8×10^{-3}	1.7×10^{-4}	1.8×10^{-7}
4	-0.0355	1.6×10^{-4}	5.0×10^{-6}	2.3×10^{-10}
5	0.0128	1.6×10^{-5}	1.8×10^{-7}	4.0×10^{-13}
6	-0.0047	1.8×10^{-6}	7.2×10^{-9}	7.9×10^{-16}
L^∞ error bound		$\le 10^{-21}$	$\le 10^{-42}$	$\le 10^{-105}$

Figure 14 shows $F_n^{(j_0)}$ for $j_0 = 2, \ldots, 6$ and $n = 1, 2, 5$. As more dominant eigenfunctions are removed, the Markov partition structure of the Gauss map becomes increasingly visible: for $j_0 \ge 4$, the peaks of $F_n^{(j_0)}$ concentrate near $x \approx 1/k$, reflecting the cylinder boundaries $I_k = [1/(k+1), 1/k]$. The amplitude decreases by roughly a factor of $|\lambda_{j_0}/\lambda_{j_0-1}| \approx 1/3$ per removed mode. Even at the

smallest amplitudes ($\sim 10^{-16}$ for $j_0 = 6$, $n = 5$), the L^∞ error bound ($\leq 10^{-105}$) remains more than 80 orders of magnitude below the signal, confirming that the plotted curves are rigorous to plotting accuracy.

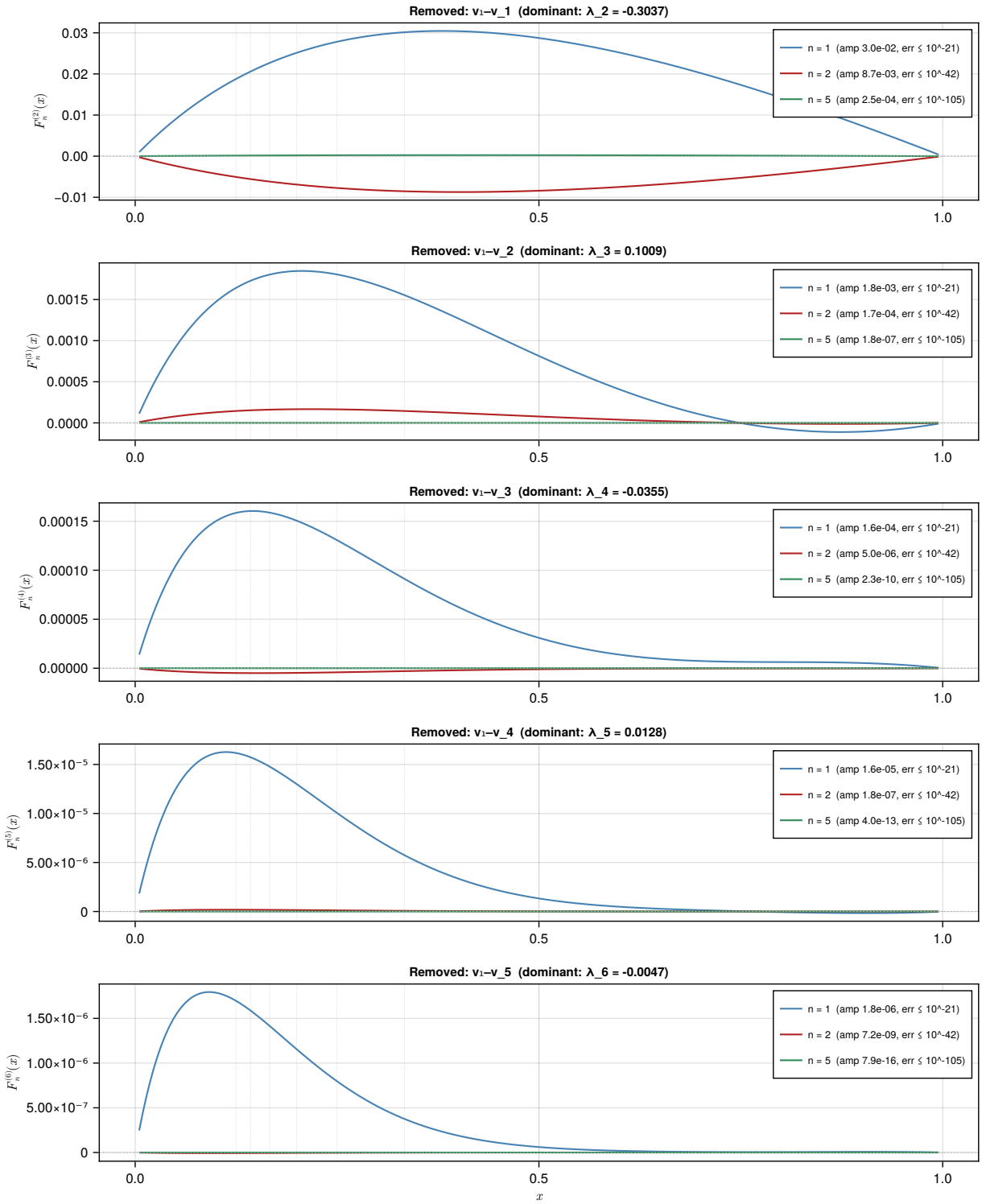

Figure 14: Integrated spectral expansion $F_n^{(j_0)}(x) = \int_0^x \sum_{j \geq j_0} \lambda_j^n \, \ell_j(\mathbf{1}) \, v_j(t) \, dt$ with the first $j_0 - 1$ eigenfunctions cumulatively removed. Each panel shows $n = 1$ (blue), $n = 2$ (red), $n = 5$ (green). Dotted vertical lines mark the Markov partition boundaries $x = 1/k$.

Eigenfunction Structure and the Markov Partition

The Gauss map $G(x) = \{1/x\}$ admits a countable Markov partition into cylinders $I_n = [1/(n+1), 1/n]$ for $n = 1, 2, 3, \ldots$. We investigate the relationship between the eigenfunctions v_j and this partition by analysing zero crossings and local extrema.

Zero crossings. The invariant density v_1 is strictly positive on $(0, 1)$, as expected. For $j \geq 2$, the eigenfunction v_j has a small number of zero crossings: odd-indexed eigenfunctions typically have 2 zeros, even-indexed have 1 or 3. The first zero of v_j sits near $x \approx 1/j$, matching the left endpoint of the cylinder I_{j-1}. For $j \gtrsim 40$, the polynomial representation in the shifted monomial basis $\{(x-1)^k\}$ develops spurious Runge-type oscillations at Float64 precision, producing artificially many zero crossings (e.g., ~ 2900 for v_{50}).

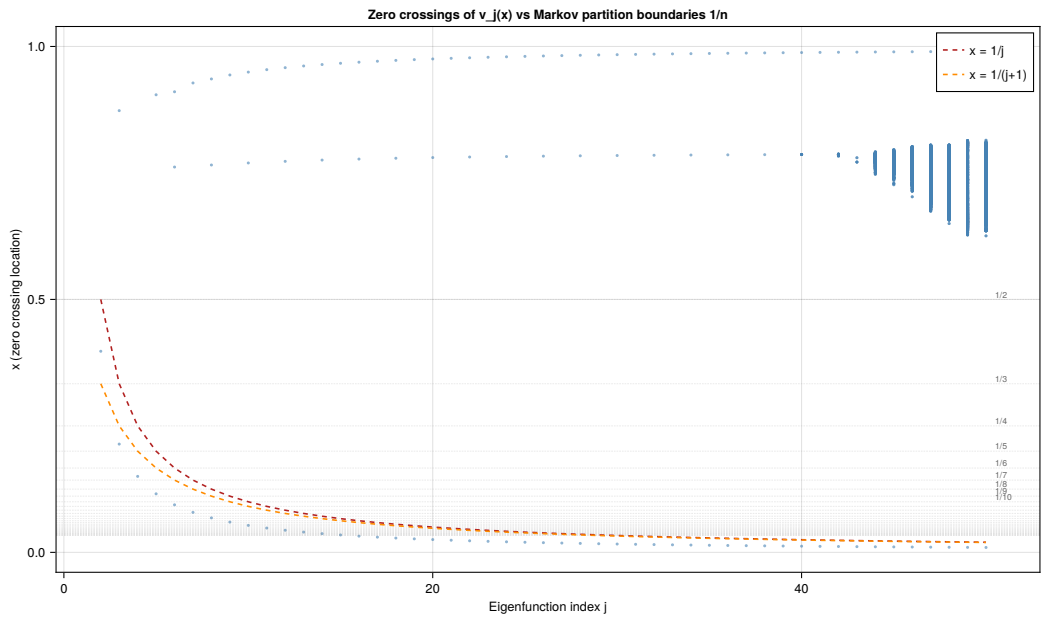

Figure 15: Zero crossing locations of $v_j(x)$ (blue dots) versus Markov partition boundaries $x = 1/n$ (dotted grey). The reference curves $x = 1/j$ (red dashed) and $x = 1/(j+1)$ (orange dashed) are shown.

Dominant local extremum. For each $j \geq 3$, the largest local extremum (in absolute value) of v_j is located near the Markov boundary $x = 1/j$:

$$x_{\max}(v_j) \approx \frac{1}{j}.$$

For $j = 3, \ldots, 24$, the extremum lies slightly above $1/j$, inside $I_{j-1} = [1/j, 1/(j-1)]$; for $j \geq 25$, it drifts slightly below $1/j$, into $I_j = [1/(j+1), 1/j]$. The relative error $|x_{\max}(v_j) - 1/j|/(1/j)$ is tabulated below for selected j:

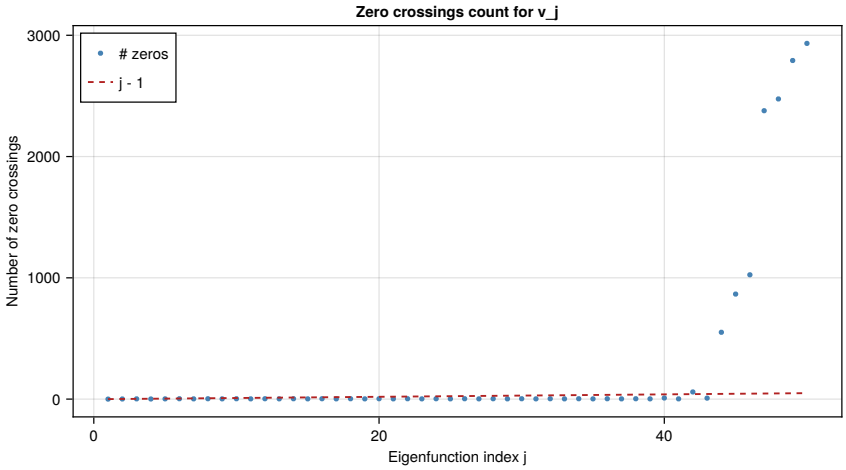

Figure 16: Number of zero crossings of v_j versus j. For $j \leq 39$ the count stays at 1–3; the explosion for $j \geq 40$ is a Runge artefact of the polynomial basis at Float64 precision.

j	$x_{\max}(v_j)$	$1/j$	rel. error	cylinder
3	0.45498	0.33333	36.5%	I_2
5	0.23258	0.20000	16.3%	I_4
10	0.10663	0.10000	6.6%	I_9
20	0.05056	0.05000	1.1%	I_{19}
30	0.03296	0.03333	1.1%	I_{30}
40	0.02439	0.02500	2.4%	I_{40}
50	0.01932	0.02000	3.4%	I_{51}

For $j \geq 10$ the mean relative error is 2.3% (std. dev. 1.5%); for $j \geq 20$ it is 1.8% (std. dev. 1.1%). Each eigenfunction is thus "tuned" to the Markov boundary $x = 1/j$, confirming that higher eigenmodes encode progressively finer partition structure.

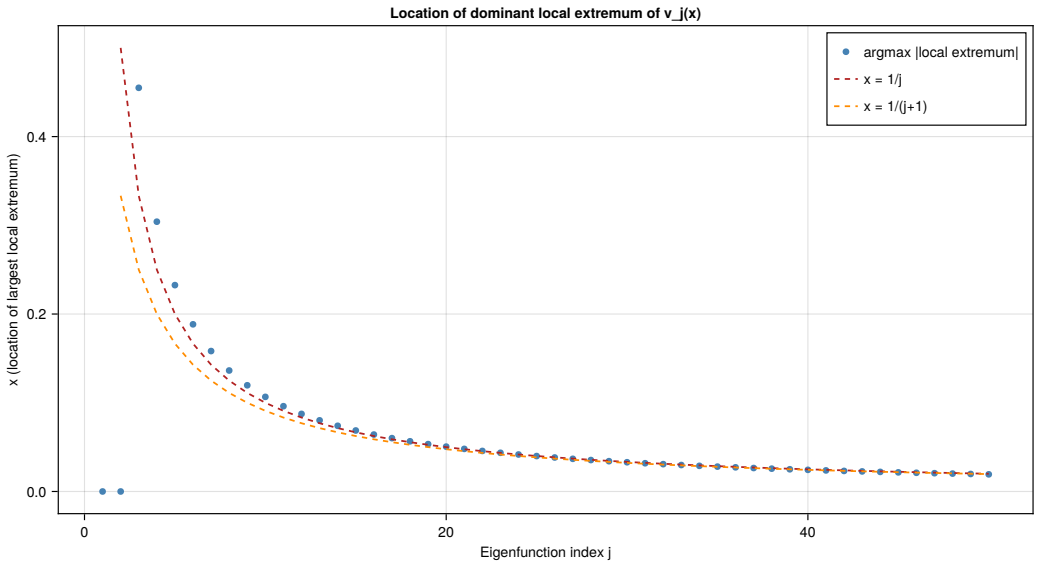

Figure 17: Location of the dominant local extremum of $v_j(x)$ (blue dots) compared with $x = 1/j$ (red dashed) and $x = 1/(j+1)$ (orange dashed). For $j \geq 10$ the extremum tracks $1/j$ with $\sim 2\%$ relative error.

Non-Normality of the Transfer Operator

The GKW transfer operator is not normal, so its Schur decomposition $A = QTQ^*$ has a non-trivial strictly upper triangular part. We quantify this non-normality using the ordered Schur form with eigenvalues sorted by decreasing $|\lambda_j|$ on the diagonal ($K = 512$, $n = 513$).

Henrici departure from normality. The Henrici number

$$\delta_H \;=\; \frac{\|T - \mathrm{diag}(T)\|_F}{\|T\|_F} \;=\; 3.82 \times 10^{-1}$$

indicates that roughly 38% of the Frobenius norm of T resides in the off-diagonal entries, confirming significant non-normality.

Coupling structure. For fixed row i, the entry T_{ij} with $j > i$ quantifies the coupling from Schur mode j (smaller $|\lambda_j|$) back into mode i (larger $|\lambda_i|$). Figure 18 shows the full 80×80 heatmap of $\log_{10} |T_{ij}|$; Figure 19 displays the row profiles $|T_{i,j}|$ for selected i.

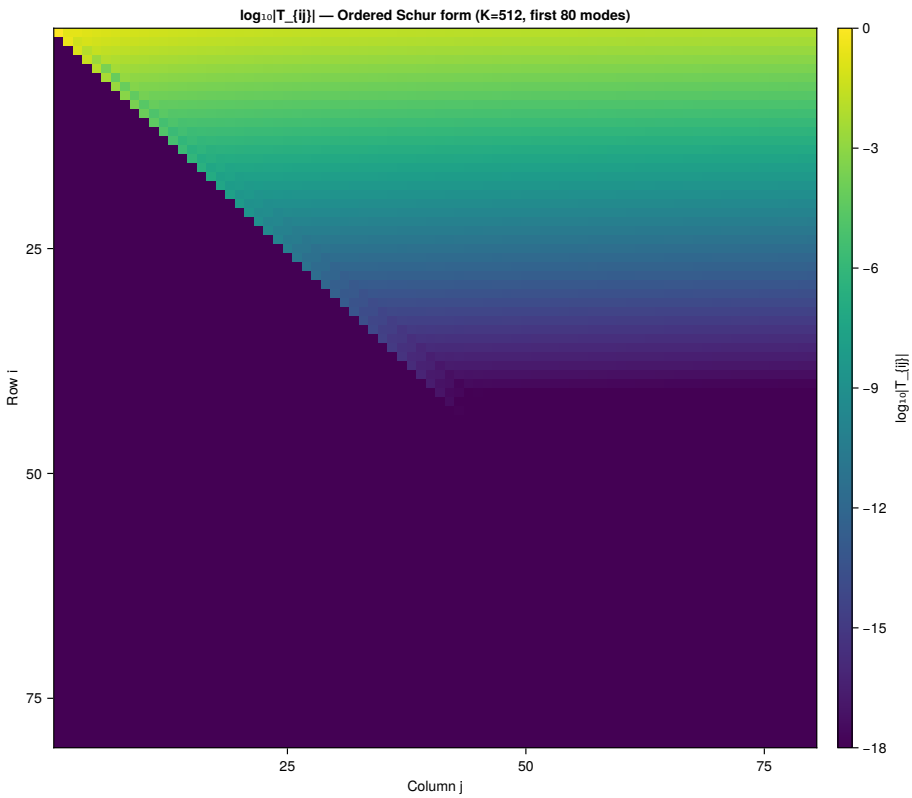

Figure 18: Heatmap of $\log_{10} |T_{ij}|$ in the ordered Schur form (eigenvalues sorted by decreasing $|\lambda|$, first 80 modes). The diagonal contains the eigenvalues; the upper triangular entries encode non-normal coupling between modes.

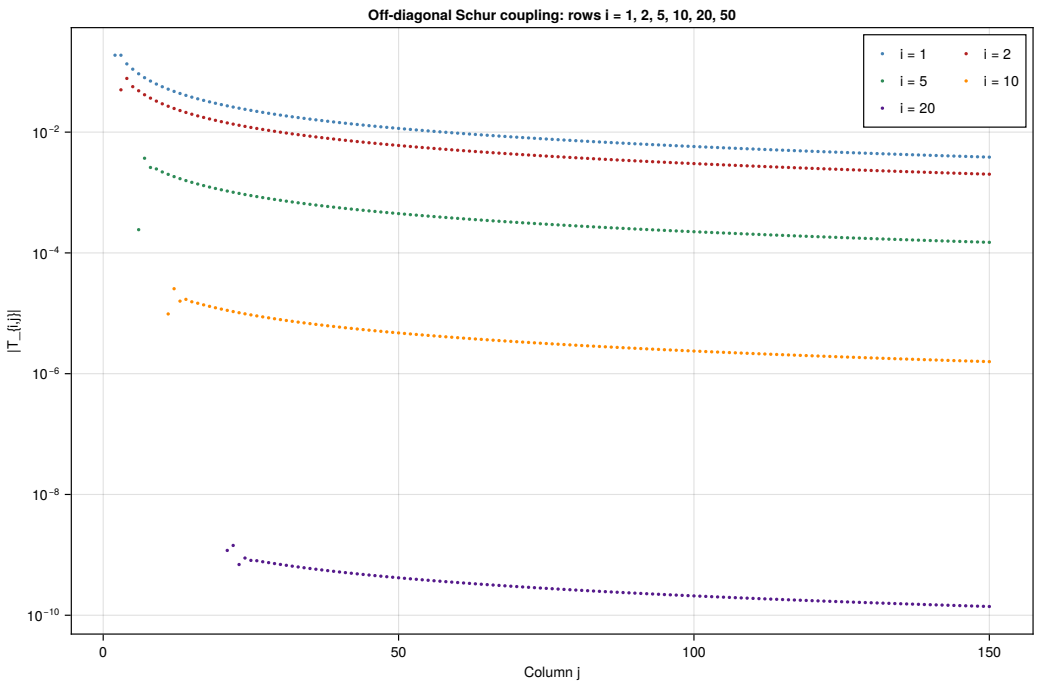

Figure 19: Off-diagonal Schur coupling $|T_{i,j}|$ for rows $i = 1, 2, 5, 10, 20, 50$ (log scale). Mode j feeds back into mode i with strength $|T_{i,j}|$.

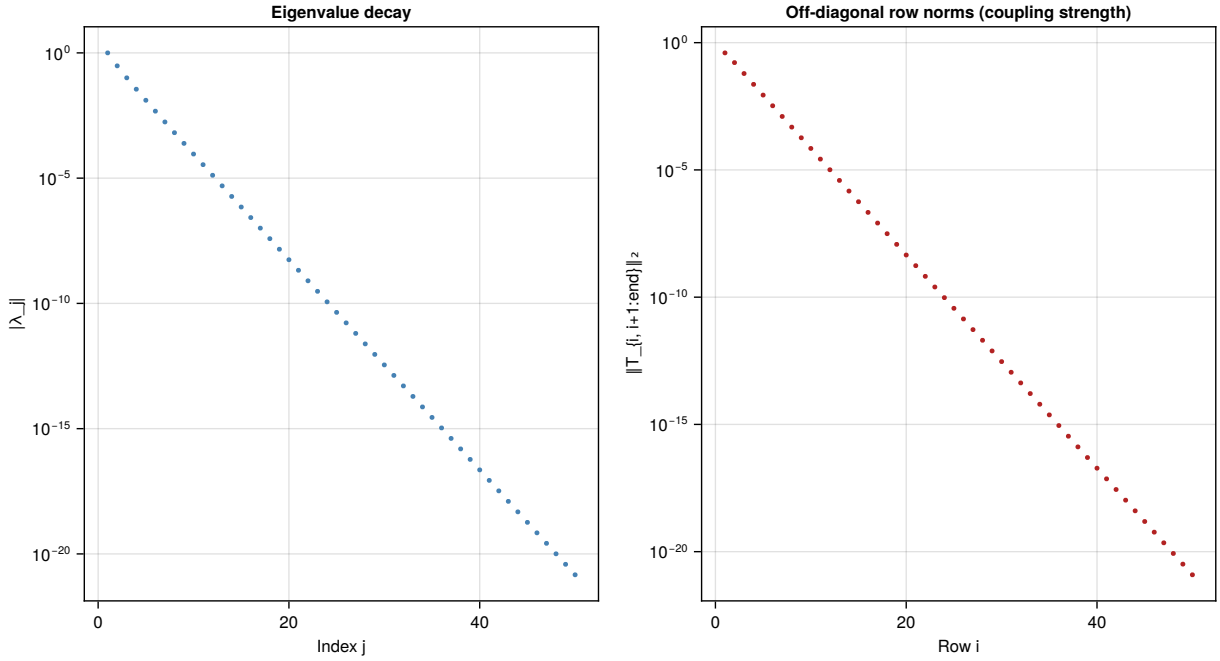

Figure 20: Left: eigenvalue decay $|\lambda_j|$ versus index j. Right: off-diagonal row norms $\|T_{i,i+1:n}\|_2$ measuring the total coupling strength from all higher modes into mode i.